

XC24R

The controller for constant current drivers

24 Channels

General Description

The XC24R is a local dimming controller featuring 24 channels to drive its companion chips, XDR series LED drivers that act as the constant current sink for Color LEDs back-lighting solution. This controller interfaces with XDR series drivers through one serial line to send and receive the data to be needed for local dimming.

The XC24R supports an SPI interface for communication with the host. Multiple XC24R devices can share the SPI bus, with each device selected by an individual Chip Select (CS) signal, allowing flexible expansion for large-scale local dimming applications.

For raising up dimming controllability, XC24R supports the FB signal to efficiently control the LED power supply voltage and two interrupt pins, nINT_FAULT to indicate the fault events like as the thermal status, the open detection, the short detection and the feedback status, and nINT_LD to indicate the room of internal buffers for dimming data.

Application

Micro RGB LED back-lighting with local dimming control

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Revision History

Date (YYYY-MM-DD)	Ver.	Editor	Contents
2026/01/26	0.10	YJE	Initial release.
2026/09/04	0.50	YS Eom JM Yu	4p Block diagram & Application circuit update. 5p Pin description VSYNCI update. 7p Pin description FBI1,2,3 Remark update if not used, 10kΩ PU 30p Figure 25 multiple XCs FBI pin pull up update
2026/09/09	0.51	YS Eom JM Yu CH Ye	3p ESDHBM : ±4KV update 5~8p Pin description PAD type update. 9p Equivalent Circuits L_Gx , H_Gx pad update. 12p AMR L_Gx, H_Gx update 27~28p Local Dimming data buffer update. 30p Fig 24 The scheme in using single XC image update 31p DAC control channels update. 32p Command packet format Id transfer Line size update. Command packet format Register write update. 66p DAC control DAC_AUTO_TYPE update.
2026/09/09	0.51	JS Ho	33p 'h00 [B6] VSYNC_RST3 update 42p 'h13 [B9] INT_SPI_PC_EN → INT_PARITY_ERR_EN 50p 'h1B [B11] LD_RD_CLK_SEL → BUF_READ_SPRD [B3] SERIALIZE_CLK_SEL → SERIAL_SPRD_EN 59p 'h3B [B7:B0] SVO_DELAY update 64p 'h41 [B9:B8] SV_NO_TYPE → SV_NO_MULT 65p 'h43 [B15] DAC_FAULT_OFF → DAC_THERMAL_OFF 72p 'h51 [B6:B0] DAC1_INT4 → DAC_INC4 75p 'h58 [B6:B0] DAC2_INT4 → DAC2_INC4 78p 'h5F [B6:B0] DAC3_INT4 → DAC3_INC4

Features

- 24 channels local dimming controller for the XD mono and RGB series LED drivers
- Flexible sizing for each channel
 - up to 31 daisied LED drivers per channel
 - up to 255 blocks dimming zone per channel
 - up to 6,120 blocks dimming zone per XC24R
- SPI interface
 - Supports SPI clock frequencies up to 30 MHz
 - Supports multiple devices using individual Chip Select (CS) signals
 - Supports packet error detection using an access counter and parity bit
- Configurable the serial communication baud-rate speed
- Internal OSC 35MHz with spread function for EMI reduction
 - Configurable MCLK, using a external MCLK or internal OSC MCLK
- Configurable the serial data out skew between each channel for EMI reduction
- Frame synchronization support
- Fault events support of drivers through the interrupt function
 - Over-temperature event
 - Open detection
 - Short detection
 - FB status for VLED power efficiency
 - V_{DD} , V_{15} OV/UV detection
 - VSYNC missing detection
- 12-bit DAC 3 channel for VLED control by FB
 - Configurable Auto-controllability
- TTL-level input range
- Supply voltage 3.0V ~ 3.6V
- 68-PIN QFN 8080 Package
- ESD protection, ESD_{HBM} (JEDEC JS-001-2017 Human Body Model) : $\pm 4KV$

Block Diagram

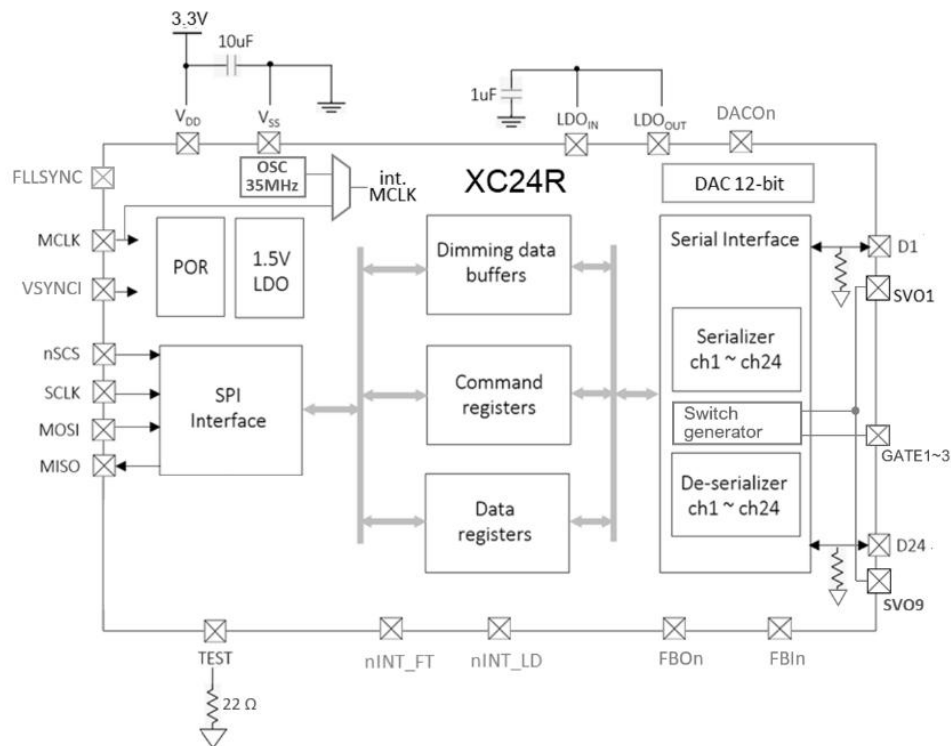


Figure 1. Block Diagram

Application Circuit

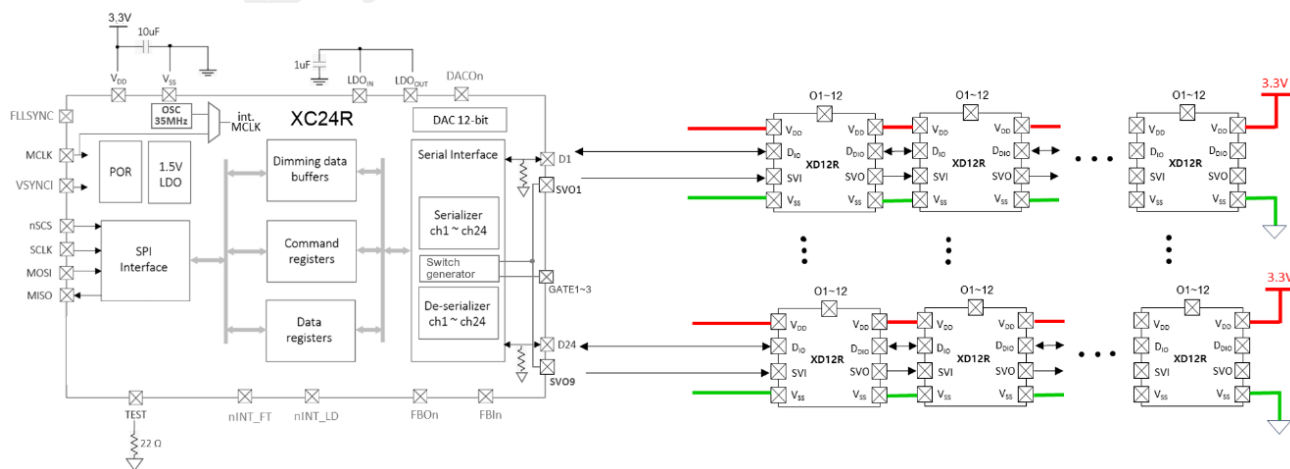


Figure 2. Application Circuit

Pin diagram (Top View)

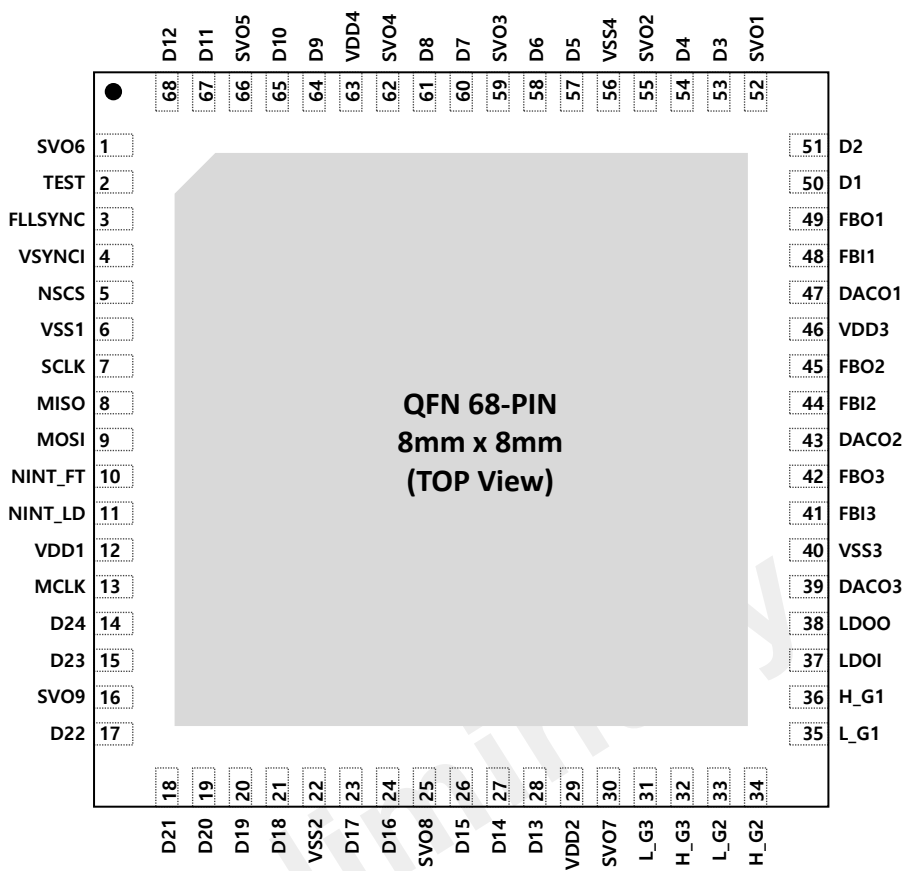


Figure 3. Pin diagram

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
SVO6	1	O	Digital	Sub Vsync. Output	
TEST	2	I	Digital	For only TEST mode	0Ω PD
FLLSYNC	3	O	Digital	Frequency Locked Loop Sync	
VSYNCI	4	I	Digital	Frame synchronization	
NSCS	5	I	Digital	SPI Chip Select	
VSS1	6	S	GND	Digital GND	
SCLK	7	I	Digital	SPI clock	
MISO	8	O	Digital	SPI MISO	

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
MOSI	9	I	Digital	SPI MOSI	
NINT_FT	10	O	Digital	Interrupt for the fault events	Open-drain, 10kΩ PU
NINT_LD	11	O	Digital	Interrupt for the dimming data room	Open-drain, 10kΩ PU
VDD1	12	S	Power	Digital VDD	
MCLK	13	I	Digital	Master Clock	if not used, connect to GND
D24	14	I/O	Digital	Data In/Out	if not used, leave to open
D23	15	I/O	Digital	Data In/Out	if not used, leave to open
SVO9	16	O	Digital	Sub Vsync. Output	
D22	17	I/O	Digital	Data In/Out	if not used, leave to open
D21	18	I/O	Digital	Data In/Out	if not used, leave to open
D20	19	I/O	Digital	Data In/Out	if not used, leave to open
D19	20	I/O	Digital	Data In/Out	if not used, leave to open
D18	21	I/O	Digital	Data In/Out	if not used, leave to open
VSS2	22	S	GND	Digital GND	
D17	23	I/O	Digital	Data In/Out	if not used, leave to open
D16	24	I/O	Digital	Data In/Out	if not used, leave to open
SVO8	25	O	Digital	Sub Vsync. Output	
D15	26	I/O	Digital	Data In/Out	if not used, leave to open
D14	27	I/O	Digital	Data In/Out	if not used, leave to open
D13	28	I/O	Digital	Data In/Out	if not used, leave to open
VDD2	29	S	Power	Digital VDD	
SVO7	30	O	Digital	Sub Vsync. Output	
L_G3	31	O	Digital	VLED3 gate control Output	
H_G3	32	O	Digital	VLED3 gate control Output	
L_G2	33	O	Digital	VLED2 gate control Output	
H_G2	34	O	Digital	VLED2 gate control Output	

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
L_G1	35	O	Digital	VLED1 gate control Output	
H_G1	36	O	Digital	VLED1 gate control Output	
LDO1	37	I	Power	LDO feedback from LDOOUT	attach 1uF
LDOO	38	O	Power	Internal LDO out to LDOIN	connect to LDOIN
DACO3	39	O	Analog	DAC for VLED control	
VSS3	40	S	GND	Digital GND	for Analog test pin
FBI3	41	I	Digital	Feedback for VLED control	in case of using multiple XCs if not used, 10kΩ PU
FBO3	42	O	Digital	Feedback for VLED control	Open-drain, 10kΩ PU
DACO2	43	O	Analog	DAC for VLED control	
FBI2	44	I	Digital	Feedback for VLED control	in case of using multiple XCs if not used, 10kΩ PU
FBO2	45	O	Digital	Feedback for VLED control	Open-drain, 10kΩ PU
VDD3	46	S	Power	Digital VDD	
DACO1	47	O	Analog	DAC for VLED control	
FBI1	48	I	Digital	Feedback for VLED control	in case of using multiple XCs if not used, 10kΩ PU
FBO1	49	O	Digital	Feedback for VLED control	Open-drain, 10kΩ PU
D1	50	I/O	Digital	Data In/Out	if not used, leave to open
D2	51	I/O	Digital	Data In/Out	if not used, leave to open
SVO1	52	O	Digital	Sub Vsync. Output	
D3	53	I/O	Digital	Data In/Out	if not used, leave to open
D4	54	I/O	Digital	Data In/Out	if not used, leave to open
SVO2	55	O	Digital	Sub Vsync. Output	
VSS4	56	S	GND	Digital GND	
D5	57	I/O	Digital	Data In/Out	if not used, leave to open
D6	58	I/O	Digital	Data In/Out	if not used, leave to open
SVO3	59	O	Digital	Sub Vsync. Output	
D7	60	I/O	Digital	Data In/Out	if not used, leave to open

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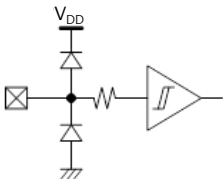
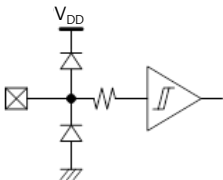
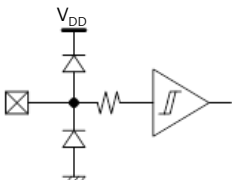
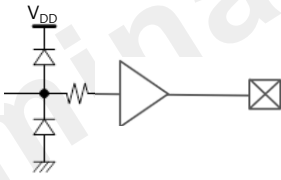
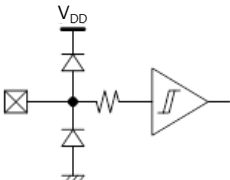
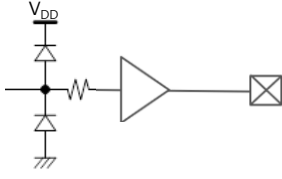
Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
D8	61	I/O	Digital	Data In/Out	if not used, leave to open
SVO4	62	O	Digital	Sub Vsync. Output	
VDD4	63	S	Power	Digital VDD	
D9	64	I/O	Digital	Data In/Out	if not used, leave to open
D10	65	I/O	Digital	Data In/Out	if not used, leave to open
SVO5	66	O	Digital	Sub Vsync. Output	
D11	67	I/O	Digital	Data In/Out	if not used, leave to open
D12	68	I/O	Digital	Data In/Out	if not used, leave to open

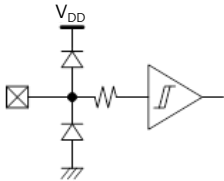
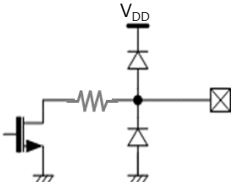
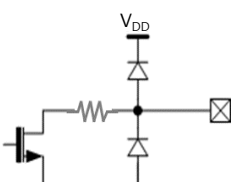
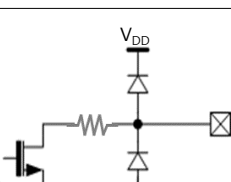
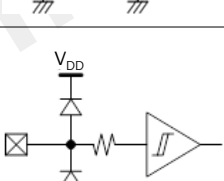
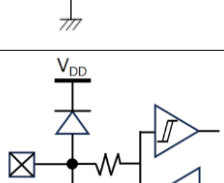
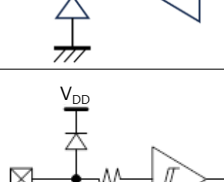
Note :
1) I = input, O = output, S = power supply.

Table 1. Pin Description

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
nSCS	I		SPI Chip Select
SCLK	I		SPI Clock Input
MOSI	I		SPI MOSI
MISO	O		SPI MISO
MCLK	I		Mater Clock
L_Gx H_Gx	O		VLED1 Gate Control

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
VSYNC	I		Frame Sync
nINT_FT	O		Interrupt for Fault (Open-drain)
nINT_LD	O		Interrupt for LD (Open-drain)
FBO	O		Feedback Output for VLED (Open-drain)
FBI	I		Feedback Input for VLED
D	I/O		Data In/Out
TEST	I		For test only

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
LDO _{IN}	I		Internal LDO In
LDO _{OUT}	O		Internal LDO Out
DACO	O		DAC Out
V _{DD}	I		Digital Power
V _{SS}	I		Digital GND

Figure 4. Equivalent Circuits of individual pins

Absolute Maximum Ratings (Ta = 25°C unless otherwise noted)

Parameter	Symbol	Value	Unit
Supply voltage	V _{DD}	-0.3 ~ 6.0	V
MCLK to GND voltage	V _{MCLK}	-0.3 ~ V _{DD} +0.5	V
VSYNC to GND voltage	V _{VSYNC}	-0.3 ~ V _{DD} +0.5	V
nSCS to GND voltage	V _{SCS}	-0.3 ~ V _{DD} +0.5	V
SCLK to GND voltage	V _{SCLK}	-0.3 ~ V _{DD} +0.5	V
MOSI to GND voltage	V _{MOSI}	-0.3 ~ V _{DD} +0.5	V
MISO to GND voltage	V _{MISO}	-0.3 ~ V _{DD} +0.5	V
L_Gx , H_Gx to GND voltage	V _{L_Gx} V _{H_Gx}	-0.3 ~ V _{DD} +0.5	V
D voltage	V _{DIO}	-0.3 ~ V _{DD} +0.5	V
Thermal Resistance			
Junction to Ambient	R _{θJA}	68 ¹⁾	°C/W
Junction to Case	R _{θJC}	10 ¹⁾	°C/W
Power Dissipation on package	P _{TOT}	1.03 ¹⁾	W
Ambient temperature	T _A	-20 ~ 85	°C
Junction temperature	T _J	-20 ~ 150	°C
Storage temperature	T _{STG}	-45 ~ 150	°C
Relative Humidity(non-condensing)	RH _{NC}	5 ~ 85	%
Moisture Sensitivity Level	MSL	Maximum Floor Life-time of 168 hr	3

Table 2. Absolute Maximum Ratings

Note :

1) Surface mounted on FR-4 substrate PC Board 1" x 1" inch 2-oz copper plate, T_A = 55°C

2) Stresses beyond the Maximum Absolute ratings may cause permanent damage to the device

3) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device.

These are stress ratings only, and do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions. Exposure to absolute- maximum-rated conditions for extended periods may affect device reliability.

ESD Ratings

Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM	ESD _{HBM}	±4000	V	Norm: JEDEC JS-001-2017
Electrostatic Discharge CDM	ESD _{CDM}	±500	V	Norm: JEDEC JESD22-C101F Oct 2013

Table 3. ESD ratings

Electrical Characteristic (V_{DD} =3.3V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply						
Operating voltage	V _{DD}		3.0	3.3	3.6	V
Operating Current	I _{DD}		-	6.95	-	mA
Quiescent current	I _S		-	2.80	-	mA
MCLK (external)						
Frequency	F _{MCLK}		1)	-	35	MHz
Duty		@ 1.5V level	45	50	55	%
Jitter	△F _{MCLK}		-1		1	%
MCLK (internal OSC)						
Frequency	F _{MCLK}		-	35	-	MHz
Duty			-	-	-	
Jitter	△F _{MCLK}		-	-	-	%
MCLK (Serial data OSC ¹⁾)						
Frequency	F _{MCLK}		-	35	-	MHz
Duty			-	-	-	
Jitter	△F _{MCLK}		-	-	-	%
Input						
nSCS Input Voltage	V _{SCSL}	Chip Select, low level	-	-	0.8	V
	V _{SCSH}	Chip Select, high level	2.0	-	V _{DD}	V
SCLK Input Voltage	V _{SCLKL}	SPI Clock, low level	-	-	0.8	V
	V _{SCLKH}	SPI Clock, high level	2.0	-	V _{DD}	V
MOSI Input Voltage	V _{MOSIL}	MOSI, low level	-	-	0.8	V
	V _{MOSIH}	MOSI, high level	2.0	-	V _{DD}	V
VSYNC Input Voltage	V _{VSYNCL}	VSYNC low level	-	-	0.8	V
	V _{VSYNCH}	VSYNC high level	2.0	-	V _{DD}	V
FBI Input Voltage	V _{VSYNCL}	FBI low level	-	-	0.8	V
	V _{VSYNCH}	FBI high level	2.0	-	V _{DD}	V
Output						
MISO Output voltage	V _{MISOL}	MISO, low level	-	-	0.5	V
	V _{MISOH}	MISO, high level	V _{DD} *0.8	-	V _{DD}	V

Note :
1) Serial data out OSC need to set faster than internal OSC for transferring data stable.

Electrical Characteristic (V_{DD}=3.3V, Ta=25.0°C unless otherwise noted)

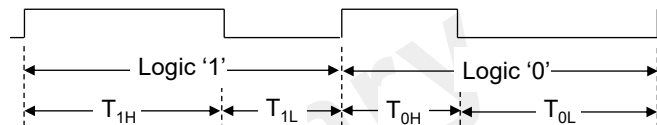
Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	

Bi-direction

D _{IO}	V _{IL} V _{IH}	Input logic low level	-	-	0.8	V
		Input logic high level	2.0	-	V _{DD}	
	V _{OL} V _{OH}	Output logic low level	-	-	0.5	V
		Output logic high level	V _{DD} *0.8	-	V _{DD}	
	R _{PD}	Internal pull-down resistance	-	20	-	KΩ

Bi-direction : Input signal decoding capability

Logical long time ²⁾	T _{1H} , T _{0L}		6	-	63	MCLK
Logical short time ²⁾	T _{1L} , T _{0H}		3	-	32	MCLK



DAC

DACO	Operating voltage	V _{DD}	3.0	3.3	3.6	V
	Resolution	Available Min value decimal 100		-	12	Bit
	Output voltage	Available Max value decimal 4,000	0.1	-	3.2	V
	Code error	△V _{out}	-	±3	±5	%

V_{DD} Ramp-up and V_{LED} Start

V _{DD} Ramp-up time	T _{VDD}	from 0 to stable 3.3V			2.0	ms
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Internal Initialization after V_{DD}, 3.3V stable

Internal Initial time	T _{INIT}	after V _{DD} , 3.3V	100			ms
Allowed 1 st access	T _{FACC}	after T _{INIT} is completed	5			ms

Table 4. DC Electrical Characteristic

Note : 1) It is recommended the minimum MCLK should not be more than twice as slow as the SCLK for stable fast-readout. If MCLK will be more than twice as slow as SCLK, then an appropriate latency is needed between the first 16 SCLKs for a command and the second 16 SCLKs for readout.

2) The conditions, T_{1H} ≥ T_{0H} * 2, T_{0L} ≥ T_{1L} * 2, (T_{1H} + T_{0H}) = (T_{0L} + T_{1L}) < 80 are normally recommended. However, the value depends on the baud-rate of the XDR..

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

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Electrical Characteristic (V_{DD} = 3.3V, Ta = 25.0°C unless otherwise noted)

Power Sequence

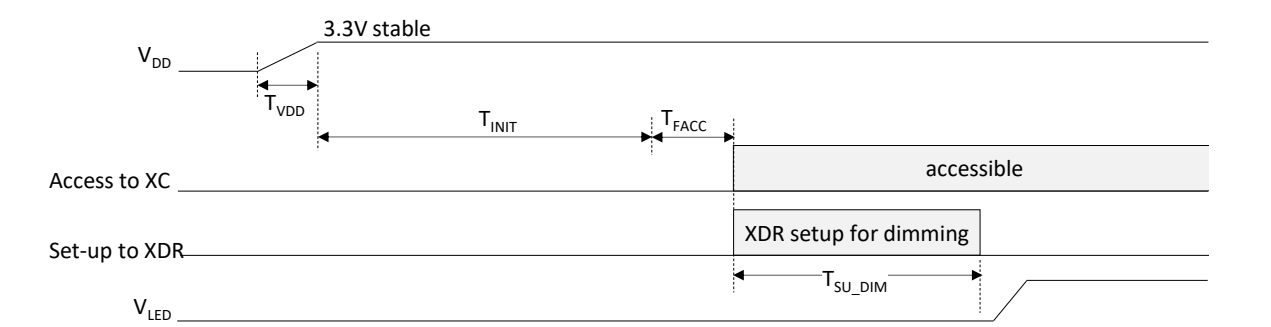


Figure 5. Power up sequence and ramp-up time

AC Characteristics (V_{DDs}=3.3V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
SCLK						
SCLK frequency	f _{CLK}		-	-	30	MHz
SCLK duty	t _{CLK_DUTY}		48	-	52	%
SCLK high time	t _{CH}		8	-	-	ns
SCLK low time	t _{CL}		8	-	-	ns
SCLK dependent time						
nSCS setup to the first rising SCLK	t _{CSS}		50	-	-	ns
nSCS hold time from the last falling SCLK	t _{CSH}		3	-	-	MCLK
nSCS high latency time	t _{CSHG}		3	-	-	MCLK
MOSI setup time	t _{DS}		5	-	-	ns
MOSI hold time	t _{DH}		2	-	-	ns
MISO delay from the rising edge SCLK	t _{DO}		-	-	12	ns
MISO high-Z from the nSCS de-assertion	t _{DOZ}		-	-	1	MCLK

Note (3) It should be careful that the t_{CO_DELAY}, t_{SKEW}, t_{OCH} and t_{DO} are accumulated by the number of daisy devices if the daisy chain connection is used. So, for the read transaction, the proper timing should be considered in daisy chain.

Table 5. AC Electrical Characteristic

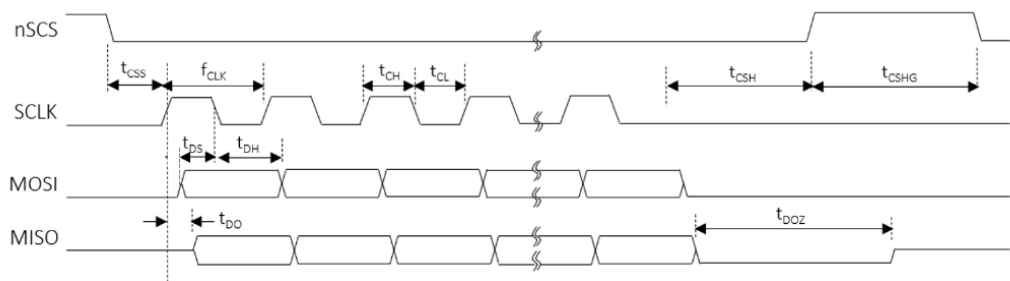


Figure 6. SPI timing sequence

Functional Description

1. SPI Interface

The local dimming data is transmitted through a Serial Peripheral Interface (SPI). The SPI is configured to operate as an SPI slave and supports clock frequencies of up to 30 MHz.

SPI Chip Select Structure

Multiple SPI slave devices are supported. Each device has an independent Chip Select (CS) signal, allowing the SPI master to communicate with each device individually.

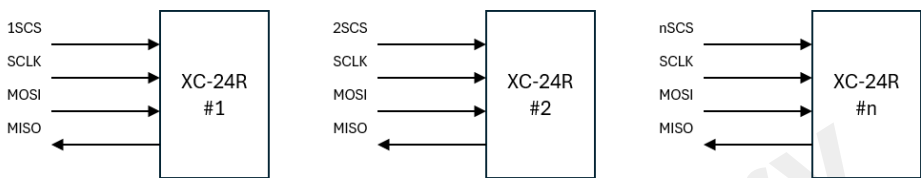


Figure 7. SPI Chip Select

SPI Protocol

The XC24R has 3 types of the SPI Interface protocol, the register-read, the register-write and the LD data-transfer. The SPI protocol's format consists of a 16-bit command and a 16-bit register data or a series of 16bit LD data following the 16-bit command.

[Command format]

	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	Command code [1:0]		Address [7:0]								Burst size [5:0]					
Register-read	0	1	address [7:0]								Burst size [5:0]					
Register-write	1	0	address [7:0]								Burst size [5:0]					

	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
LD-transfer	1	1	don't care				Line size [9:0]									

[Data format]

	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	Data [15:0]															
Register-RW	Register data [15:0]															
LD transfer	16 bits / block															

Functional Description

SPI LD transfer sequence

The SPI sequence for the LD transfer is as 12~16bits. The LD width depends on the LD_WIDTH value of the CHANNEL ENABLE 2 register. If the LD_WIDTH = 2'b01, then the LD data size of one block zone transferred through the SPI becomes 16 bits. Even if the transmitted width is 16 bits, the valid bits are actually 11 bits to 16 bits depending on LD_WIDTH setting.

LD_WIDTH [1:0]	LD width	LD format	Valid bits
2'b01	12 bits	PWM 12-bit	PWM 12-bit
2'b10	14 bits	PWM 14-bit	PWM 14-bit
2'b11	16 bits	LD 16-bit	LD 16-bit

Table 6. LD data size according to LD_WIDTH

The burst size unit means the amount of LD data corresponding to one line and the one line of LD data corresponds to the total LD data of all activated channels, and the number of active channel is determined by the CH_SIZE value of the ENABLE CHANNEL 2 register. Here, the CH_SIZE value should be equal to the number of activated channels by the CHx_EN of the ENABLE CHANNEL 1/2 registers.

Burst size unit = 1-line LD data = CH_SIZE * LD data size

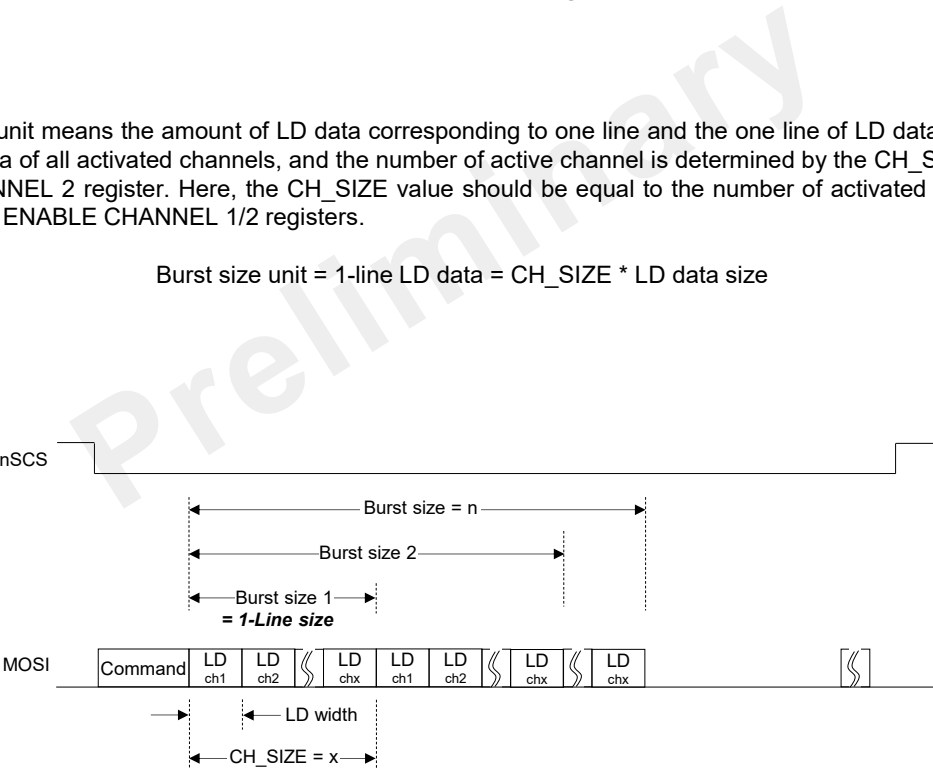


Figure 10. The SPI LD transfer

Functional Description

2. Serial Interface

The XC24R has a total of 24 channels for serial daisy connections with a total of 31 XDR devices. Each channel can transmit the data to the XDR and receive the data from the XDR through one line daisy to the XDR devices. The XC24R has a total of 6 instructions for transferring the data to the XDR and 3 instructions for receiving the data from the XDR.

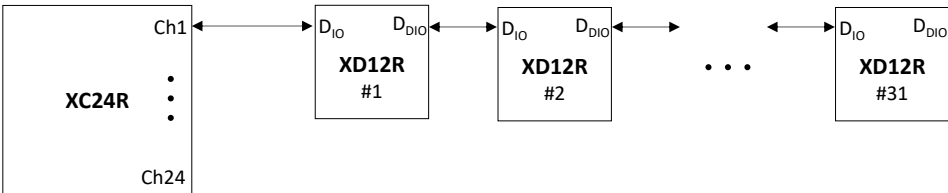


Figure 11. The daisy scheme

The commands are listed as shown in the Table 7. and described in more detail below.

Instruction	Transaction direction		Description
	Command	Data	
Global-write	XCR → XDR	XCR → XDR	Global-write command the same register data between the XDR devices
Local-write	XCR → XDR	XCR → XDR	Local-write command the different register data between the XDR devices
LD-transfer	XCR → XDR	XCR → XDR	Local Dimming data-transfer command to the XDR devices
ID-gen	XCR → XDR	-	ID generation command of the XDR devices in daisy each channel
Sync-gen	XCR → XDR	-	Sync-generation command to the XDR devices
Sync-auto-gen	XCR → XDR	-	Automatically sync-generation command sync on the XDR devices
Local-read	XCR → XDR	XDR → XCR	Local-read command to the XDR devices and read-back from the XDR devices
Fault-read	XCR → XDR	XDR → XCR	Fault-read command to the XDR devices and read-back from the XDR devices
Fault-auto-read	XCR → XDR	XDR → XCR	Automatically fault-read command to the XDR devices and read-back from the XDR devices

Table 7. Instruction list

Functional Description

ID-generation

The ID-GEN COMMAND register is used to set a unique ID for each device and each device's ID helps identifying when each device should react on a read-back operation. Therefore, after POR this command should be transacted at the first time for proper operation of the XDR devices before transacting other commands. If the START bit of ID-GEN COMMAND register is set, then all activated channels begin issuing the command signals to all XDRs daisied in each channel and the command signal streams repeat continuously for the number of daisied devices.

The number of daisied devices is determined by the DAISY SIZE register of each channel. After the command signal streams are completed, all devices will have their own ID respectively. The ID is numbered in opposite direction of daisied line. It is careful that the writing data is transferred from the first located device, the last number ID, to the last located device, the first number ID, in daisied line but the reading data is received from the last located device to the first located device sequentially.

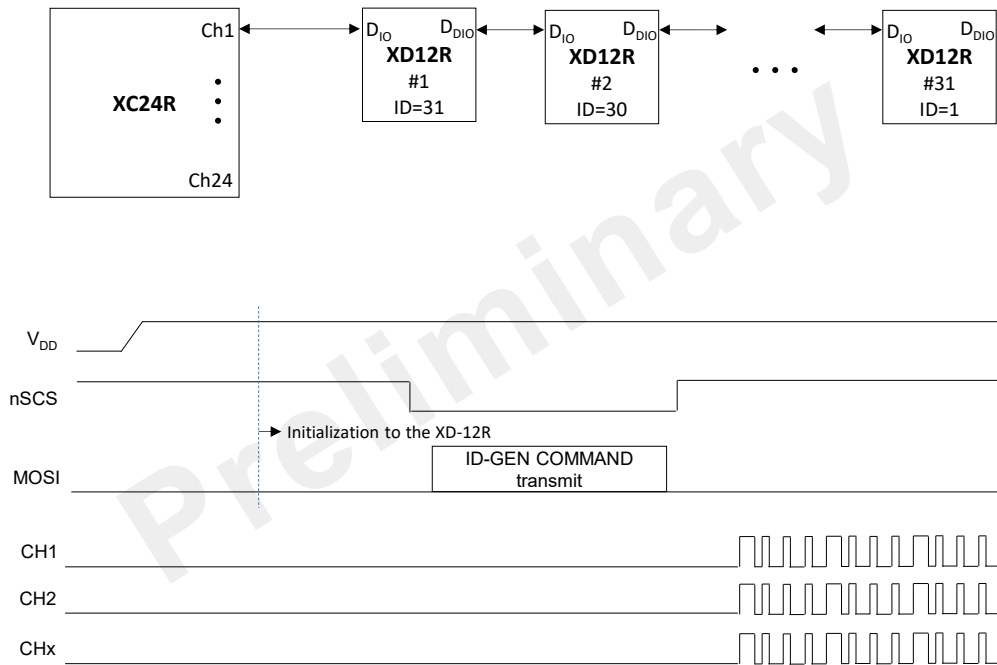


Figure 12. The sequence of ID-GEN command

Functional Description

Global-write

The Global-write command is used to write the register data to the XDR only if all XDR devices daisied to a channel of the XC24R have the same register data. If the register data to be written to the XDRs are different each other, at that time the Local-write command should be used. For executing the Global-write command properly, the GLOBAL-WRITE DATA register should be firstly written with desired value and then the START bit of the GLOBAL-WRITE COMMAND register should be set with desired address in order to begin transacting this command.

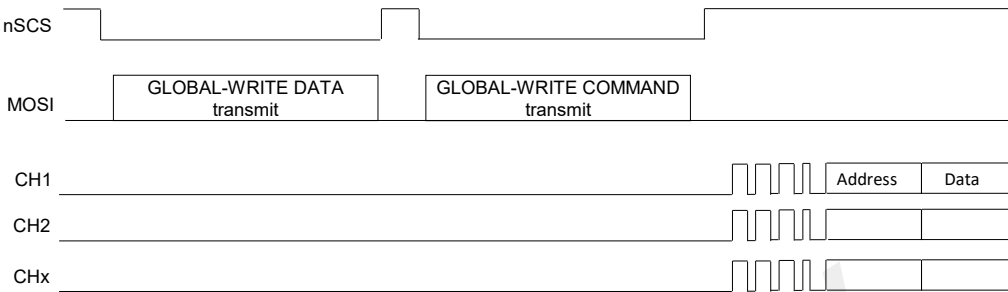


Figure 13. The sequence of Global-write command

The GLOBAL-WRITE COMMAND register consists of 6 bits of ADDRESS field and 1 bit of START. The ADDRESS is the register address of the XDR to be accessed. If the START bit is set with the ADDRESS together, the Global writing command begins issuing simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XDRs. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31.

Functional Description

Local-write

The Local-write command is used to write the register data to the XDR in case that all XDR devices daisied to a channel of the XC24R have the different register data each other. If the register data to be written to the XDR are same, at that time the Global-write command is more useful for fast transaction time. For proper execution of the Local-write command, the LOCAL-RW DATA registers should be firstly written with desired value and then, the START bit of the LOCAL-WRITE COMMAND register should be set with desired address for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local - write command execution is completed or before the next Local-write command will start again.

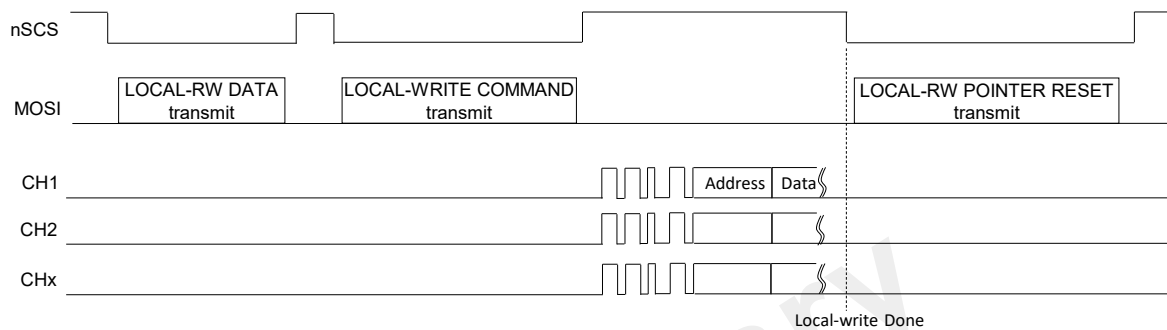


Figure 14. The sequence of Local-write command

The LOCAL-WRITE COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XDR to be accessed. If the START bit is set with the ADDRESS together, the Local-write command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XDR. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time. The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. The first data register in the port is mapped to the first located XD12R in daisied order and the second data register is mapped to the second located XD, and so on. Although the data registers in a port are a total 16, the XC24R can support over 16 daisied devices in a channel by using the LOCAL-WR TRANSFER POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-WR TRANSFER DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data remain in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must continue to fill up before they are empty. A value greater than 16 means that the buffer has been overwritten, so you must wait until the pointer value is less than 16.

CH_SEG[1:0]	Channel to be activated	LOCAL-RW DATA 1 ~ 16							
		PORT1	PORT2	PORT3	PORT4	PORT5	PORT6	PORT7	PORT8
2'b00	Ch 1 ~ Ch 8	Ch 1	Ch 2	Ch 3	Ch 4	Ch 5	Ch 6	Ch 7	Ch 8
2'b01	Ch 9 ~ Ch 16	Ch 9	Ch 10	Ch 11	Ch 12	Ch 13	Ch 14	Ch 15	Ch 16
2'b10	Ch 17 ~ Ch 24	Ch 17	Ch 18	Ch 19	Ch 20	Ch 21	Ch 22	Ch 23	Ch 24

Table 8. The LOCAL-RW DATA mapping according to the CH_SEG

Functional Description

Local-RW DATA structure for access

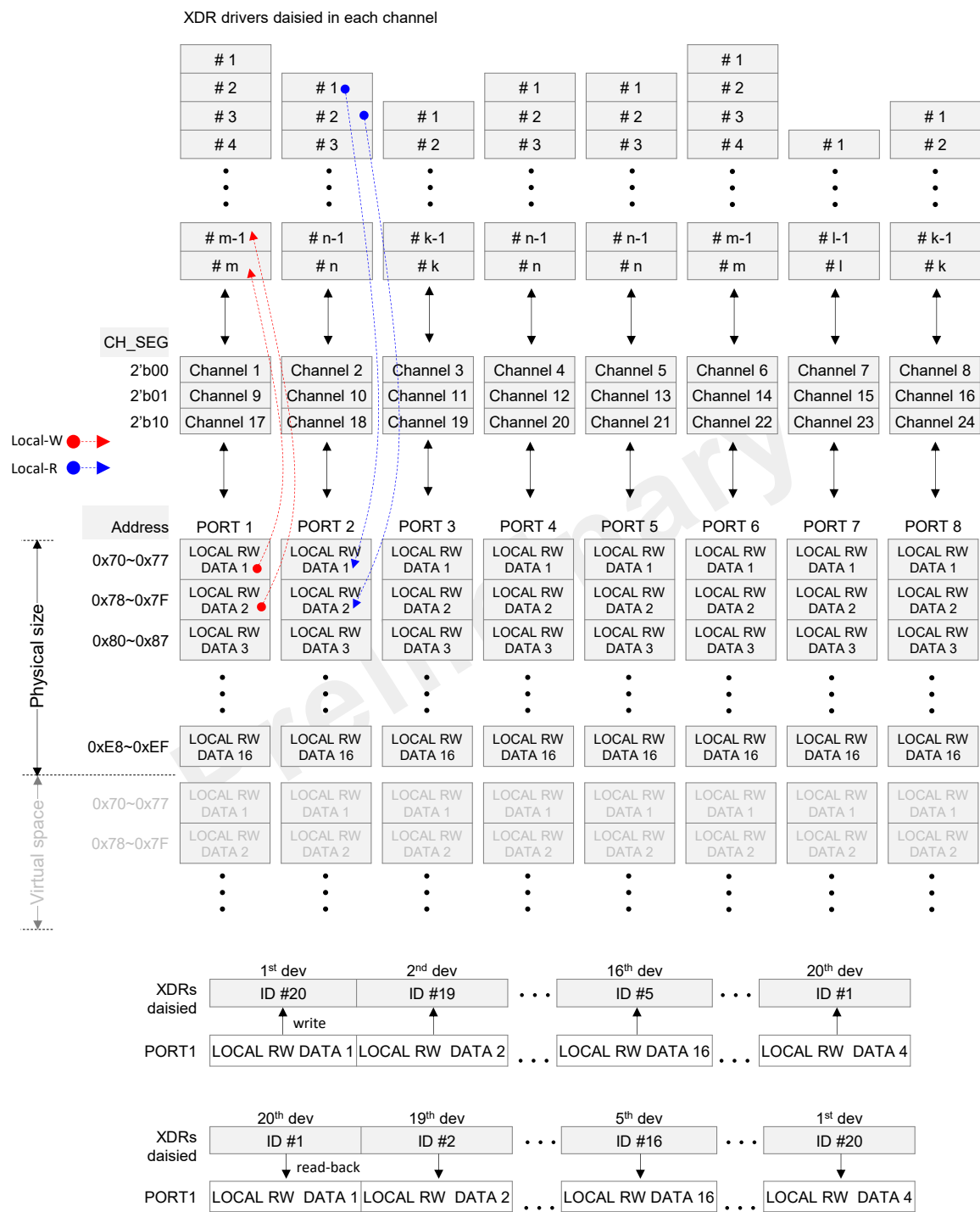


Figure 15. LOCAL-RW DATA mapping structure and data flow

Functional Description

Sync-generation

The Sync-generation command is used to synchronize the frame sync time for all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the SYNC-GEN COMMAND register and the other is by setting the SYNC_AUTO_EN bit of the COMMAND AUTO ENABLE register. If the SYNC_AUTO_EN bit is set, the Sync-generation command is automatically executed at the moment of receiving the VSYNC input. It is recommended to use this function, but it should be careful that The synchronization time of the XDR devices may be as late as the number of XDR devices than the time the VSYNC was received, so the VSYNC should arrive as soon as it is late. This command streams repeat continuously for the number of daisied devices in each channel.

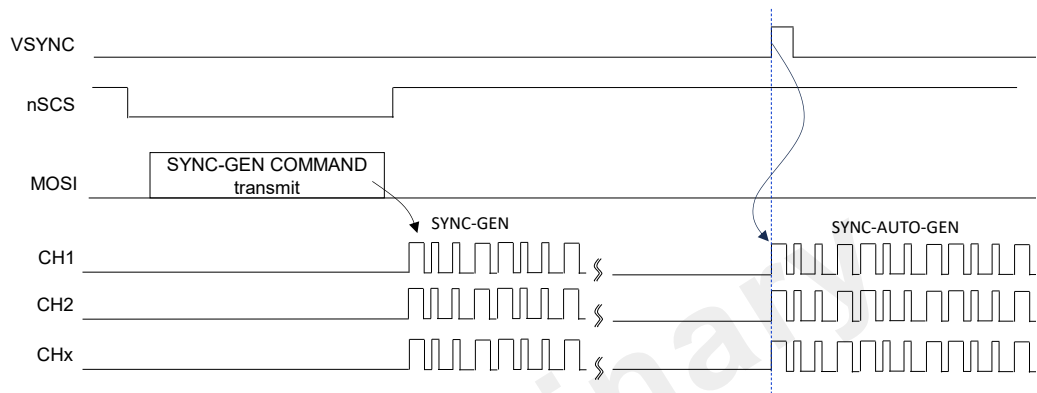


Figure 16. The sequence of SYNC-GEN command

Fault-read

The Fault-read command is used to read-out the fault information of all daisied devices simultaneously. This command execution can be done by two methods, the one is by setting the START bit of the FAULT-GEN COMMAND register and the other is by setting the FAULT_AUTO_EN bit of the COMMAND AUTO ENABLE register. If the FAULT_AUTO_EN bit is set, the Fault-read command is automatically executed after the completion of the Sync auto-gen command at the moment of receiving the VSYNC input. The fault data to be read-back from all XDR devices in each channel is stored into the corresponding channel's GLOBAL FAULT-READ DATA register, respectively.

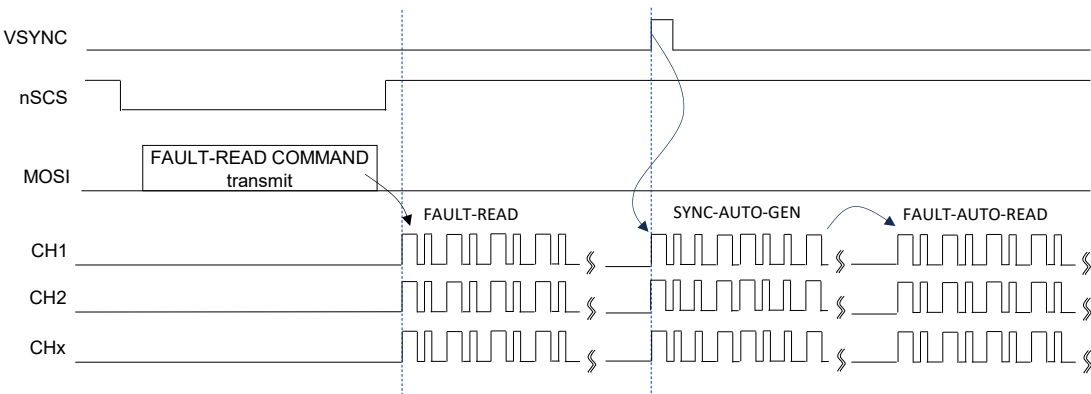


Figure 17. The sequence of FAULT-READ command

Functional Description

Local-read

The Local-read command is used to read the register data from all XDR devices daisied in a channel of the XC24R. Unlike the Local-write or Global-write command, the Local-read command can be used to read the register data of the XDR devices regardless of whether the data is the same or different. The data to be read-out from the XDR devices are stored into the LOCAL-RW DATA registers sequentially like the Figure 15. The START bit of the LOCAL-READ COMMAND register should be set with desired address to be read-out for beginning this command. It is careful that the LOCAL-RW POINTER RESET register must be set necessarily to 1 for the next transaction after the current Local read command execution is completed or before the next Local-read command will start again.

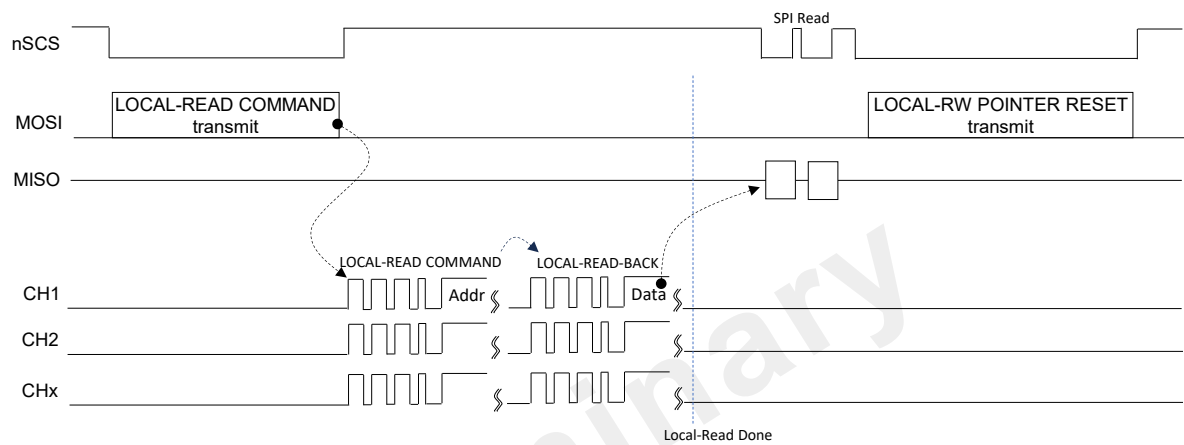


Figure 18. The sequence of Local-read command

The LOCAL-READ COMMAND register consists of ADDRESS 6 bits, CH_SEG 2 bits and START 1 bit. The ADDRESS is the register address of the XDR to be accessed. If the START bit is set with the ADDRESS together, the Local-read command begins the execution simultaneously in all activated channels, and the command streams repeat continuously for the number of daisied XDRs. The number of daisied devices can be determined by the DAISY SIZE register of each channel. The max number of daisied devices per channel is 31. The channel segment bits, CH_SEG, are used to decide on which channels are activated for this command, because this command can be performed 8 channels at a time.

The LOCAL-RW DATA registers are divided into a total of 8 port group and each port has a total of 16 registers. Unlike the Local-write command, the first data register in the port is mapped to the last located XDR in daisied order and the second data register is mapped to the second located XDR from the back, and so on. Although the data registers in a port are a total 16, the XC24R can support over 16 daisied devices in a channel by using the LOCAL-READ RECEIVE POINTER and LOCAL-RW DIFFERENCE POINTER. The LOCAL-RD RECEIVE DIFF POINTER of the LOCAL-RW DIFFERENCE POINTER shows how many data are filled up in the LOCAL-RW DATA buffers. A value of 0 means that the LOCAL-RW DATA buffers are empty, so you must wait for the buffers to be refilled, and if the value is greater than 16, it means that the LOCAL-RW DATA buffers have been overwritten, so you must drag out the buffers before they are overflowed. Refer to the Table 8. and the Figure 15. for the mapping scheme.

Functional Description

Sequence of Local Dimming data transfer

The Local Dimming data of the next frame should be transmitted to the XC24R through the SPI and the XC24R should complete to transfer the LD data to the XDR devices before assertion of the next frame sync. And then, the LD data will be started dimming-out by the XDR devices at the next frame sync. The LD_WR_POINTER of the LD WR POINTER register will be increased one by one in line unit whenever the LD data are written to the XC24R LD data buffers through the SPI and see the Figure 10. for the amount of one line LD. The LD_RD_POINTER will be increased one by one whenever the XC24R transmits the amount of one LD data to the XDR devices. The LD_DIFF_POINTER is a value of the LD_WR_POINTER minus the LD_RD_POINTER, so the value is useful to check how many LD data remains in the internal LD buffers.

If the LD_TARNS_START_POINTER value becomes equal to the LD_DIFF_POINTER, then the XC24R starts sending the LD data to the XDR devices automatically until the LD_DIFF_POINTER value becomes to 0.

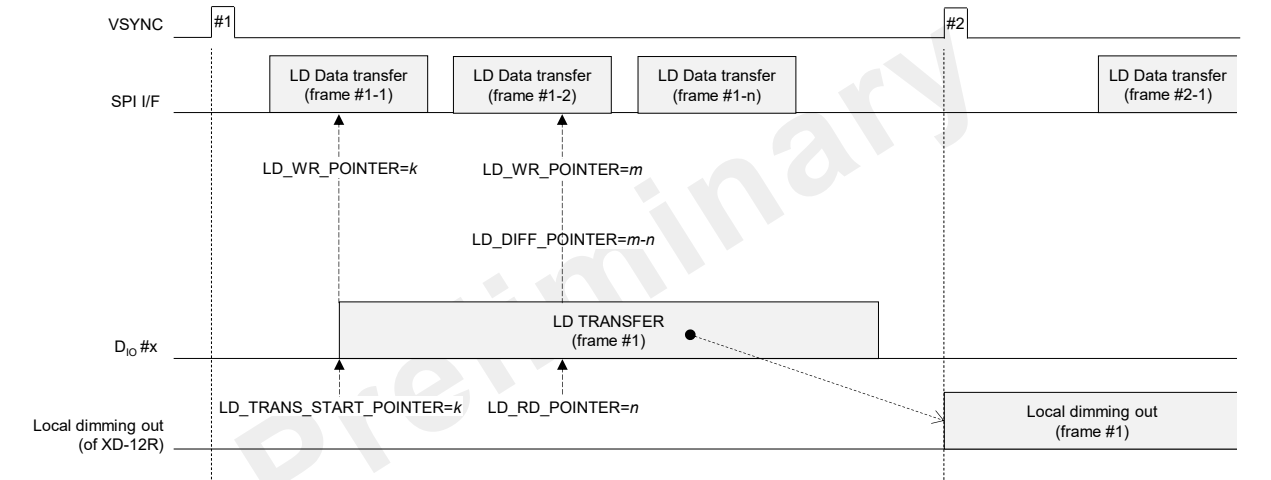


Figure 19. The sequence of Local dimming Data transfer

Functional Description

Local Dimming data buffer

The total number of Local Dimming data buffers is 54 per channel and 1 buffer can contain one block data which consists of PWM data. The size of PWM data depends on the LD_WIDTH of CHANNEL ENABLE 2 register and refer to previous chapter for more details. Even if the physical buffer size is limited to 54, the XC24R can support to transfer up to 255 blocks by using the LD WRITE POINTER, LD READ POINTER and LD DIFFERENCE POINTER. You can continue to send the LD data so that the LD DIFFERENCE POINTER does not become 0. The number of daisy devices of each channel can be different from each other by separately setting the DAISY SIZE register of each channel. The below shows the scheme of LD buffers and the mapping to the daisy device. For more information refer to the example on the next page.

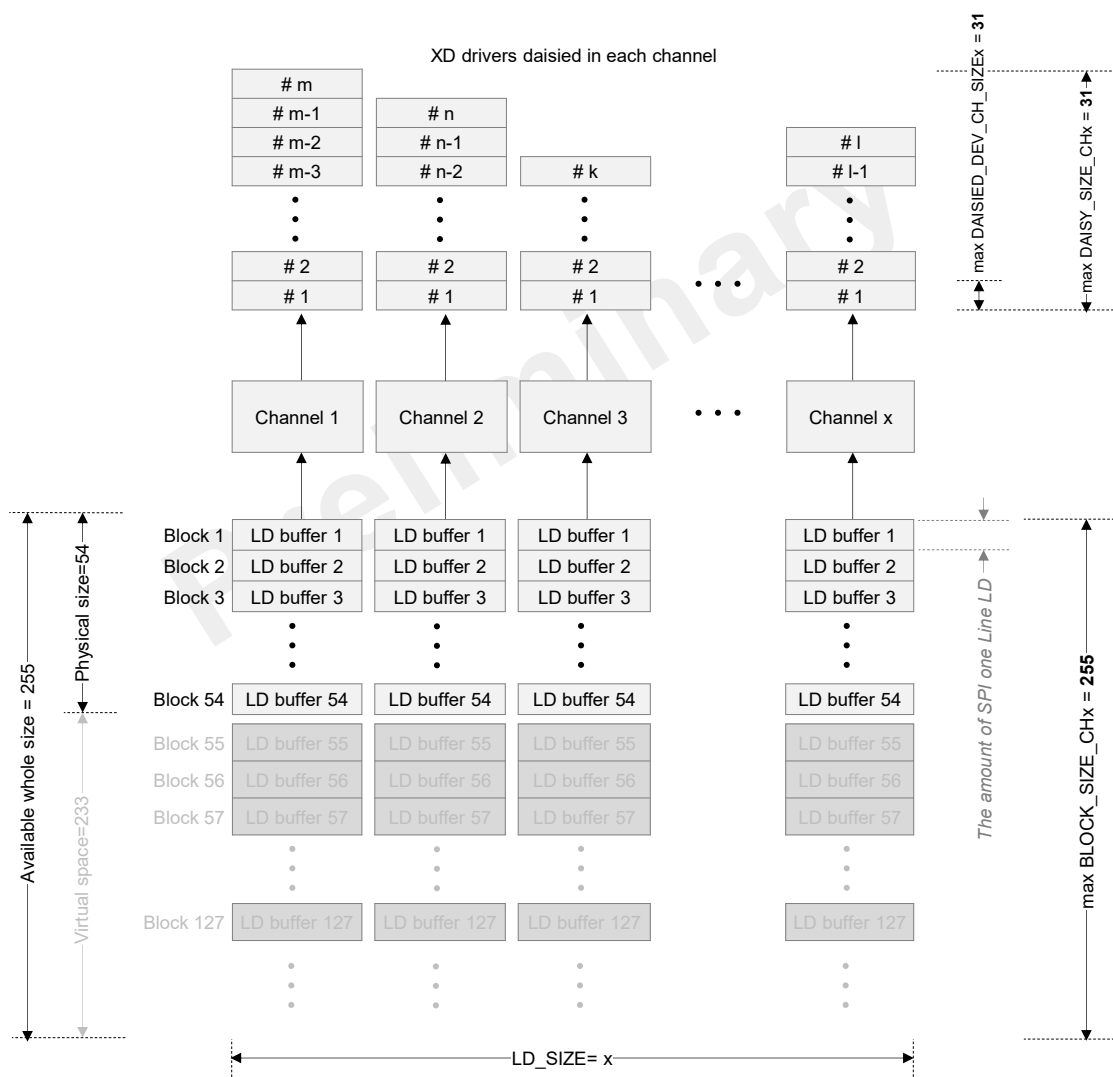


Figure 20. The scheme of Local dimming Data buffers

Functional Description

Local Dimming data transfer

The below example shows how to set properly the registers relevant to the Local Dimming data transfer. The CH_SIZE of the CHANNEL ENABLE 2 register determines how many channels among a total of 24 channels of XC24R will be activated, and if 20 channels will be used, then the LD_SIZE is 0x14 and available channels become from the channel 1 to channel 20, so it should be careful that the order of channels to be selected must be a direction that sequentially increases from 1 to 24. The DAISY_SIZE_CHx of the DAISY SIZE registers determines how many XDR devices will be daisy in the channel x, and if 10 XDR devices are daisy in a channel x, then the DAISY_SIZE_CHx should be set to 0xA. The DAISIED_DEV_CH_SIZE determines how many channels of the XDR will be used, so if the value is 0x4, the XDR devices daisy in a channel x will capture only 4 LD data and then bypass the flowing LD data streams to the next XDR devices and the remainder of the multiples of 4 be delivered lastly to a last XDR device. It should be careful that all devices except for the last device must have the same value of DAISIED_DEV_CH_SIZE. The BLOCK_SIZE_CHx determines the total number of LD data to be transferred to a channel x.

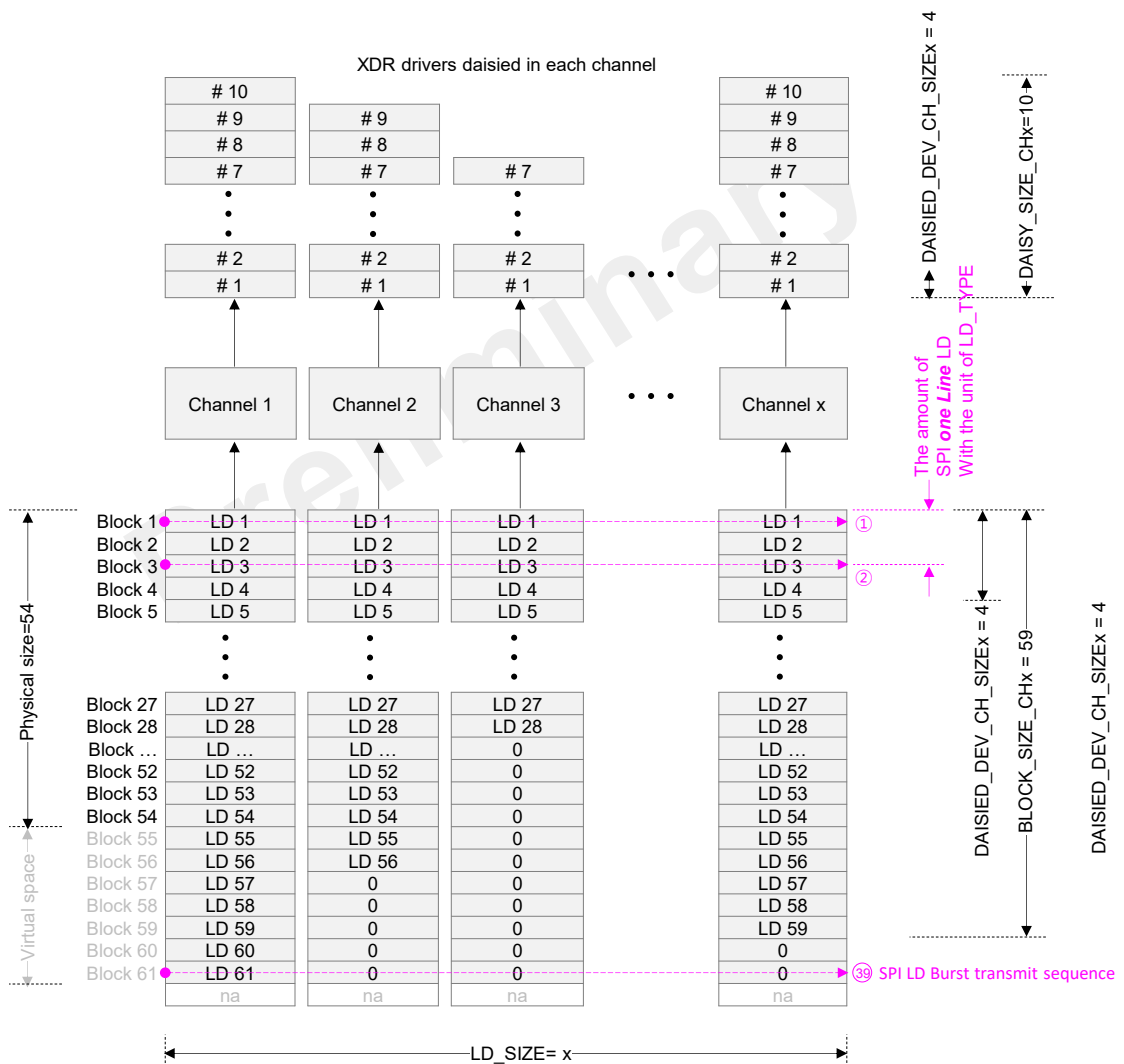


Figure 21. The example of Local dimming Data-transfer

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Functional Description

Spread clock scheme of Serializer

The internal all logics are operated with an internal MCLK clock except for the 24 Serializers. All serializers uses a Serializer clock which is configured to spread clock for reduction of EMI. All Serializers have 2 modes to reduce the EMI radiation, one is to use the spread clock, and the other is to use the output skew function between each Serializer's output. In addition, it is possible to use 2 modes at the same time. The spread amount is showed in the Table 9. below. The clock to all Serializers can be configured through 2 bits, the OSC_SPREAD_EN and the SERIALIZER_CLK_SEL. All Serializers' output skew between each Serializer's output can be configured by the SERIALIZER_SKEW_EN bit. The skew is different from all Serializers.

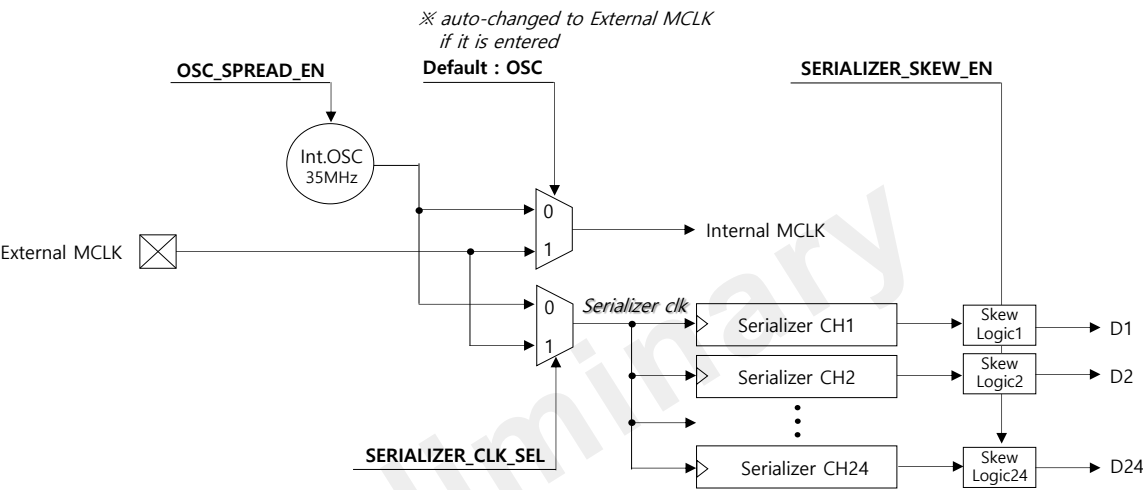


Figure 22. The scheme of the spread clock

Spread range [%]	SPREAD_GAIN [2:0]			
	0	1	2	3
	Δ55.4kHz	Δ112kHz	Δ227kHz	Δ456kHz

※ Codes 4,5,6 and 7 are not recommended for use

Table 9. The configuration of the spread range

Functional Description

The FB connection for using several XCs

The XC24R supports an internal FB status which indicates whether the VLED is enough or not. The internal FB status can be got from the readout information through the FAULT-(AUTO)-READ command. The internal FB status is driven on the FBO pin to inform other XCs of the own FB status. Instead of the internal FB status from the command, the XCs can use directly an external FB signals from the XDR devices to control the VLED like below Figure 24. The internal FB status has a value which is latched for one frame, so if the XDR will sense the FB assertion condition at least once for one frame, the readout information from the command will contain the momorized status latched for one frame. If multiple XCs are used, it needs to decide which XC is used as a master to control the VLED. Several examples are shown below, and the proper scheme can be chosen according to applications along with the DAC_FB_MODE bit.

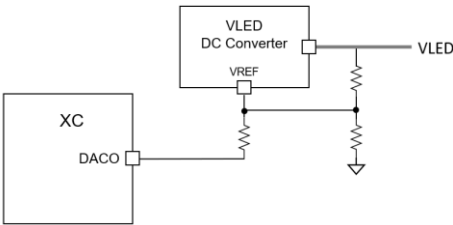


Figure 23. The DACO connection to the VLED DC Converter's VREF

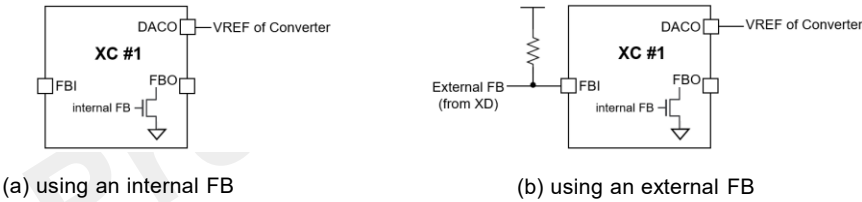


Figure 24. The scheme in using single XC

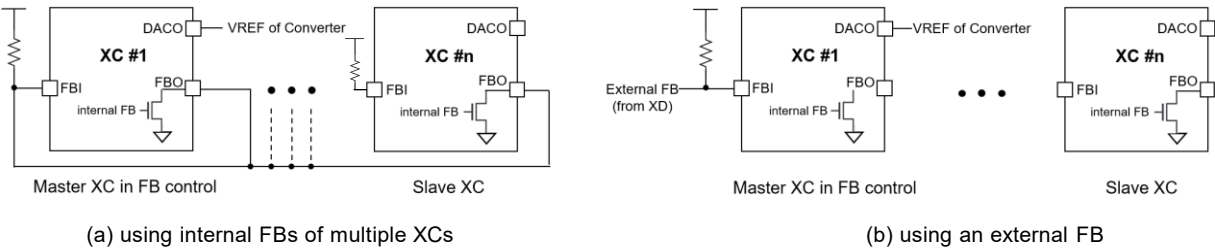


Figure 25. The scheme in using multiple XCs

Functional Description

DAC control for adjusting VLED

The XC24R has three DAC channels to adjust the VLED for efficient power consumption. The DAC out can be directly connected to a reference input of the VLED converter. The DAC has a 12-bit resolution at the max 3.3V level and can be configured to 2 modes according to the DAC_AUTO bit, the MANUAL mode or the AUTO control. In MANUAL mode, the value written to the CURR_TGT_DAC register will be output directly at incoming VSYNC signal. In AUTO mode, the DAC out is automatically controlled according to several registers' set values. The major difference between the AUTO and MANUAL mode is that the AUTO mode uses the FB status to control efficiently in order to minimize the power consumption. The proper value to be written to the CURR_TGT_DAC can be obtained by calculating of the sum of all LEDs' V_F in room temperature plus the minimum required V_{HD} of XDR. In case of low or high ambient temperature, the DAC out will be react rapidly according to the FB status. The operating conditions in the AUTO mode can be set through several registers. Refer to the register's description part for details along with Figure 27. below.

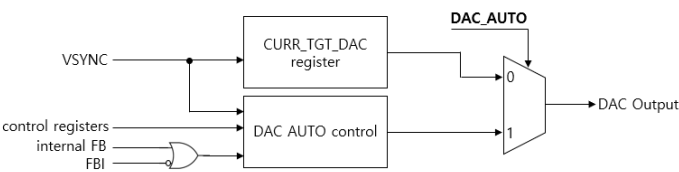


Figure 26. The scheme of the DAC output

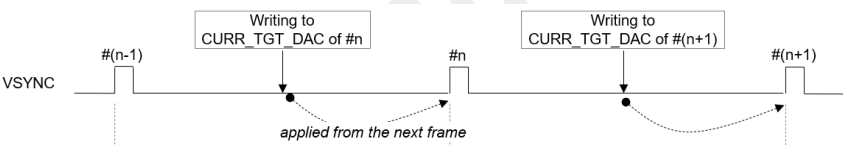


Figure 27. The effective timing of writing the CURR_TGT_DAC

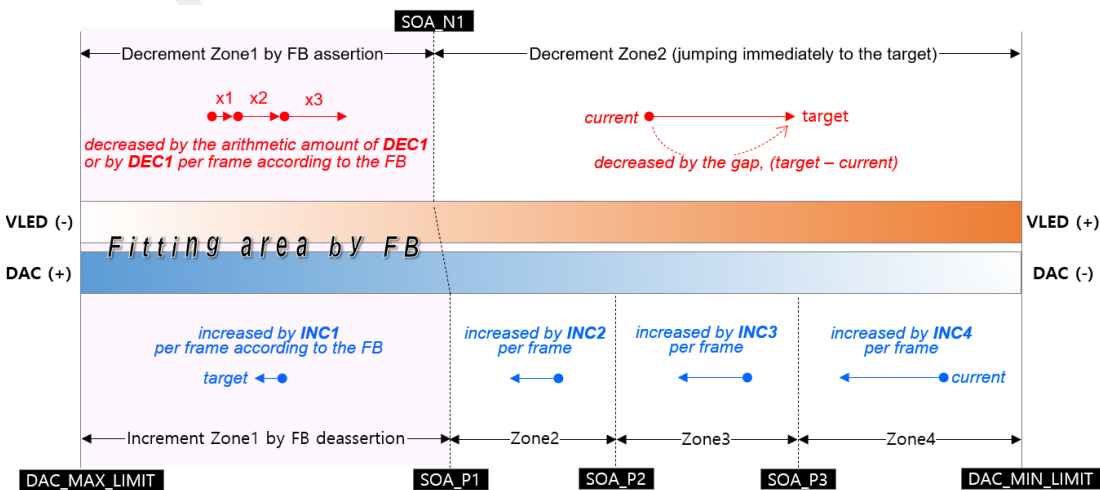


Figure 28. The scheme of the DAC AUTO control mode

3. Register Map

The SPI interface protocol consists of two segments, the command 16-bit followed by the register's data 16-bit or a series of local dimming data.
Do not write undefined register values without consulting the product supplier or vendor, as this may cause a malfunction.

Command packet format

	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	Command code [1:0]		Address [7:0]								Burst size [5:0]					
Register-read	0	1	address [7:0]								Burst size [5:0]					
Register-write	1	0	address [7:0]								Burst size [5:0]					
	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
LD-transfer	1	1	don't care				Line size [9:0]									

Bits	Fields	Descriptions
15:14	COMMAND CODE	2'b00 : No operation. 2'b01 : Register Read. 2'b10 : Register Write. 2'b11 : Local dimming data transfer.
13:6	START ADDRESS	The start address of registers to access.
5:0	BURST SIZE	Amount of data to be transmitted or received. In case of accessing the registers, the number means how many registers will be accessed for one time. In case of transmitting the LD data, the number means how many lines' LD data will be transmitted for one time, where one line means the number of enabled channels. If 20 channels are activated and the LINE SIZE is 10, and then the total amount of LD data to be transmitted becomes 20 * 10 * LD width/block [bit].

Data packet format

B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
DATA [15:0]															
Bits	Fields	Descriptions													
15:0	DATA [15:0]	In case of COMMAND CODE : 2'b01 A register's data to read. 2'b10 A register's data to write. 2'b11 A series of the LD, local dimming data. Be careful that the data size and width to transfer the local dimming data are different from the register access. Refer to previous SPI protocol													

SOFT RESET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h00	0x0030	x	x	x	x	x	x	x	x	x	VSYNC_RST3	VSYNC_RST2	VSYNC_RST1	x	RST3	RST2	RST1

Bits	Fields	Descriptions
6	VSYNC_RST3	update
5	VSYNC_RST2	If this bit is set, all de-serializers' fsm are reset at VSYNC rising. (Normally, no need to use this bit.)
4	VSYNC_RST1	If this bit is set, all serializers' fsm are reset at VSYNC rising. (Normally, no need to use this bit.)
2	RST3	Reset to all types of buffer, the frame buffers, the global and local r/w registers and the global fault read registers. Auto-cleared. (Normally, no need to use this bit.)
1	RST2	Reset to all serializers' and de-serializers' state machine including the timeout error flag, Auto-cleared. (Normally, no need to use this bit.)
0	RST1	Reset to all logics, Auto-cleared (No need to clear after set) It needs to set this bit after POR to ensure XC24R reset again.

GLOBAL-WRITE COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h01	0x0000	ENABLE	x	x	x	x	x	x	x	x	x						ADDR [5:0]

Bits	Fields	Descriptions
15	ENABLE	0 No operation. 1 Start executing the global-write command with the GLOBAL WRITE DATA register, together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the GLOBAL WRITE DATA register with desired value before setting this bit. The global-writing can be used in case that the target registers' values of all XDR daisy-chained devices are all the same. If the values are different from each other, then the local-write command should be used instead of the global-write command.
5:0	ADDR	The same target address of all XDR daisied devices to write.

LOCAL-WRITE COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h02	0x000	ENABLE	x	x	x	x	x	CH_SEG [1:0]		x	x	ADDR [5:0]					

Bits	Fields	Descriptions
15	ENABLE	0 No operation. 1 Start executing the local-write command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-write command is used in case that the target registers' values of all XDR daisy-chained devices are different from each other.
9:8	CH_SEG	These bits are used to designate which channels are activated for this command. 2'b00 The channels 1 to 8 are activated. 2'b01 The channels 9 to 16 are activated. 2'b10 The channels 17 to 24 are activated. 2'b11 Not allocated. Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG [1:0].
5:0	ADDR	The same target address of all XDR daisied devices to write

LOCAL-READ COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h03	0x0000	ENABLE	x	x	x	x	x	CH_SEG [1:0]		x	x	ADDR [5:0]					

Bits	Fields	Descriptions
15	ENABLE	0 No operation. 1 Start executing the local-read command with the LOCAL RW DATA registers together, and this bit is auto-cleared after completing this execution. For correct operation, It needs to set the LOCAL RW DATA registers with desired values before setting this bit. The local-read command is used to read out the target registers' values of all XDR daisied devices.
9:8	CH_SEG	These bits are used to designate which channels are activated for this command 2'b00 : The channels 1 to 8 are activated. 2'b01 : The channels 9 to 16 are activated. 2'b10 : The channels 17 to 24 are activated. 2'b11 : Not allocated. Refer to previous chapter. how to allocate the LOCAL RW DATA registers according to CH_SEG.
5:0	ADDR	The same target address of all XDR daisied devices to read out.

ID-GEN COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h04	0x0000	ENABLE	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bits	Fields	Descriptions
15	ENABLE	0 No operation. 1 Start executing the id-generating command to all XDR daisied devices and this bit is auto-cleared after completing this execution. All XDR daisied devices connected in each channel will have their own ID number after completing the id-generating command. This command is needed to read-out the XDR drivers' registers data correctly in sequential, so, this command should be executed before executing the local-reading or the fault-reading command. It is recommended that the id-generating command should be executed firstly on system initialization.

FAULT-READ COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h05	0x0000	ENABLE	x	x	x	x	x	x	x	x	x	x	x	x	x	FT_MODE[1:0]	

Bits	Fields	Descriptions				
15	ENABLE	0	No operation.			
		1	Start executing the fault-reading command to all XDR daisied devices to monitor the fault event like as the thermal occurrence, the short status, the open status and the fb status of the XDR devices and this bit is auto-cleared after completing this execution. This command can be also auto-executed at the VSYNC rising edge if the FAULT_ AUTO_EN bit in the COMMAND AUTO ENABLE register is set.			
1:0	FT_MODE	Interrupt source selector for bit2~5 at address 'h17.				
		ft_mode	bit5	bit4	bit3	bit2
		0	int_thermal_source	int_short_source	int_open_source	int_fb_source1
		1	int_fault_source	int_fb_source3	int_fb_source2	int_fb_source1
		2	int_fault_source	int_open_source	int_fb_source2	int_fb_source1
	int_fb_source	This bit shows the FB status from all XDRs daisied to the active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.				
	int_open_source	This bit shows the OPEN status from all XDRs daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.				
	int_short_source	This bit shows the SHORT status from all XDRs daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.				

Bits	Fields	Descriptions
	int_thermal_source	This bit shows the THERMAL status from all XDRs daisied to active channels after the FAULT-(AUTO)-READ execution. This bit value will be kept the current level until newly updated from a next FAULT-(AUTO)-READ execution even if this register has been read-out.
	int_fault_source	If this bit is set, the nINT_FAULT pin will be asserted to the active-low. This bit represents several interrupt sources, INT_FB_SRC, INT_OPEN_SRC, INT_SHORT_SRC, INT_TML_SRC, INT_LOCAL_REC_FAIL_SRC, INT_FAULT_AUTO_REC_FAIL_SRC, INT_FAULT_REC_FAIL_SRC, INT_TIMEOUT_ERR_SRC, INT_PARITY_ERR_SRC, INT_ACC_CNT_ERR_SRC and INT_OVUV_ERR_SRC if their own interrupt enable bits are set. If the corresponding interrupt enable bit is set and the interrupt source is asserted, then, this bit will be asserted through nINT_FAULT pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out.

LD-TRANSFER COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h06	0x0000	ENABLE	x	x	x	x	x	x	x	x	x	x	x	x			LD_TYPE[2:0]

Bits	Fields		Descriptions
15	ENABLE	0	No operation.
		1	Start executing the local dimming data transfer command to all XDR daisied devices in order to transfer the LD data in burst from the XC24R to the XDR drivers and this bit is auto-cleared after the completion. This command can be also auto-executed immediately after the amount of LD data to be transferred to the XC24R through the SPI interface is reached to the value of the LD_RD_START_POINTER of the LD START POINTER/TH register.
2:0	LD_TYPE		'h1 : 1 LD type : This type is for mono 'h2 : 2 LD type : This type is for 2-color mode 'h3 : 3 LD type : This type is for RGB 3-color 'h4 : 4 LD type : This type is for RGBC or RGBM 4-color 'h6 : 6 LD type : This type is for RGBCMW 6-color LD_TYPE is multiplied LD data number to transfer LD data to XDR device. The color switching signal which is SVO, the number of it should be controlled by SV_NO_TYPE on address'h41. The value of SV_NO_TYPE is also number of colors which is connected XDR channel, but these two values doesn't have to be the same value. Because it can be set only 2 color switching in the 3 colors LED assembled XDR.

SYNC-GEN COMMAND

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h07	0x0001	ENABLE	x	x	x	x	x	x	x	x	x	x	x	x	x	SYNC_MODE [1:0]	

Bits	Fields	Descriptions	
15	ENABLE	0	No operation.
		1	Start executing the sync-gen command to all XDR daisied devices in order to make the frame synchronization to the drivers and this bit is auto-cleared after this execution. This command can be also auto-executed at the VSYNC rising edge, if the SYNC_ AUTO_EN bit in the COMMAND AUTO ENABLE register is set. It is recommended that the SYNC_AUTO_EN bit is set for good local dimming.
1:0	SYNC_MODE	SVO signal mode selector 'h1 : svo only (need to turn on vsync command) 'h2 : vo signal add on svo signal 'h3 : rso signal only	

COMMAND AUTO ENABLE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h08	0x0111	x	x	x	x	x	x	x	TIME_OUT_EN	x	x	x	FAULT_AUTO_EN	x	x	x	SYNC_AUTO_EN

Bits	Fields		Descriptions
8	TIME_OUT_EN	0 1	No operation. Enable time out counter after sending read command to XDR, this responding wait timer will be expired after counting as many as the number of address 'h1F.
4	FAULT_AUTO_EN	0 1	No operation. Start executing the fault-reading command automatically. If this bit is set, the fault-reading command is auto-executed in two conditions. The first is occurred at the time of VSYNC rising edge and the second condition is occurred at the time of expiration of the FAULT AUTO READ TIMER register for the number of occurrence of the FAULT AUTO READ EVENT register.
0	SYNC_AUTO_EN	0 1	No operation. Start executing the sync-gen command automatically. If this bit is set, the sync-gen command is auto-executed at the time of VSYNC rising edge.

LD-WRITE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h09	Read only	x	x	x	x	x	x	x	LD_WR_POINTER [8:0]								

Bits	Fields	Descriptions
8:0	LD_WR_POINTER	This register value represents the number of times the local dimming data are written into the internal LD buffers through the SPI interface. The number increases one by one whenever the local dimming data corresponding to (the number of enabled channels * one LD data width) are transferred to the internal LD buffers. This pointer value is reset at the VSYNC rising.

LD-READ POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0A	Read only	x	x	x	x	x	x	x	LD_RD_POINTER [8:0]								

Bits	Fields	Descriptions
8:0	LD_RD_POINTER	This register value represents the number of times the local dimming data are read-out to transfer the data from the internal LD buffers to the XDR drivers. The number increases one by one whenever the local dimming data corresponding to the one block LD data of the XDR drivers are read-out. This pointer value is reset at the VSYNC rising.

LD-DIFFERENCE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0B	Read only	x	x	x	x	x	x	x	x	x	x	LD_DIFF_POINTER [5:0]					

Bits	Fields	Descriptions
5:0	LD_DIFF_POINTER	This value is useful to monitor how much the LD data remains in the internal LD buffers for checking overrun or underrun. = LD_WR_POINTER – LD_RD_POINTER

LD-TRANSFER START POINTER / THRESHOLD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0C	0x0000	x	x	LD_TRANS_START_POINTER [5:0]						INT_LD_SIGN	x	LD_DIFF_THRESHOLD [5:0]					

Bits	Fields	Descriptions
13:8	LD_TRANS_START_POINTER	The LD_TRANS_START_POINTER determines when the transmit from the internal LD buffers to the XDR drivers will begin. If the value is set to 8, then the transmit will begin as soon as the number of times the data through the SPI interface becomes 8. Once the transmit is stated, the transaction will be continued until the number of transmit reaches to the BLOCK_SIZE value of the BLOCK_SIZE registers.
7	INT_LD_SIGN	Don't set this bit.
5:0	LD_DIFF_THRESHOLD	The LD_DIFF_THRESHOLD is used to set in which condition the nINT_LD output signal will be issued to the host. Whenever the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESHOLD, then the nINT_LD signal will be issued. This feature helps to check how much LD data remains in the internal LD buffer for monitoring overruns or underruns of the LD data.

LOCAL-WR TRANSFER POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0D	Read only	x	x	LOCAL_WR_OUT_POINTER [5:0]						x	x	LOCAL_WR_TRANS_POINTER [5:0]					

Bits	Fields	Descriptions
13:8	LOCAL_WR_OUT_POINTER	This register value represents the number of times the data from the LOCAL RW DATA registers are read-out to write the data to the XDR drivers' registers. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers.
5:0	LOCAL_WR_TRANS_POINTER	This register value represents the number of times the data to be transferred to the XDR drivers are written to the LOCAL RW DATA registers through the SPI interface. The number increased one by one whenever the data are written to the LOCAL RW DATA registers.

LOCAL-RD RECEIVE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0E	Read only	x	x	LOCAL_RD_OUT_POINTER [5:0]						x	x	LOCAL_RD_REC_POINTER [5:0]					

Bits	Fields	Descriptions
13:8	LOCAL_RD_OUT_POINTER	This register value represents the number of times the data transferred from the XDR drivers are read out from the LOCAL RW DATA registers for transferring the data through the SPI interface. The number increases one by one whenever the data are read-out from the LOCAL RW DATA registers.
5:0	LOCAL_RD_REC_POINTER	This register value represents the number of times the data to be transferred through the SPI interface are written from the XDR drivers to the LOCAL RW DATA registers. The number increased one by one whenever the data are written to the LOCAL RW DATA registers.

LOCAL-RW DIFFERENCE POINTER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0F	Read only	x	x	LOCAL_RD_DIFF_POINTER [5:0]						x	x	LOCAL_WR_DIFF_POINTER [5:0]					

Bits	Fields	Descriptions
13:8	LOCAL_RD_DIFF_POINTER	This value is useful to monitor how much the data from the XDR drivers remains in the LOCAL RW DATA registers for checking overrun or underrun. = LOCAL_RD_REC_POINTER – LOCAL_RD_OUT_POINTER
5:0	LOCAL_WR_DIFF_POINTER	This value is useful to monitor how much the data to be transferred to the XDR remains in the LOCAL RW DATA registers for checking overrun or underrun. = LOCAL_WR_TRANS_POINTER – LOCAL_WR_OUT_POINTER

LOCAL-RW POINTER RESET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8		B7	B6	B5	B4	B3	B2	B1	B0	
'h10	0x0000	x	x	x	x	x	x	x	LOCAL_RD_POINTER_RST		x	x	x	x	x	x	x	LOCAL_WR_POINTER_RST	

Bits		Fields	Descriptions	
8	LOCAL_RD_POINTER_RST		0	No operation
			1	The LOCAL_RD_OUT_POINTER and the LOCAL_RD_REC_POINTER are reset. This bit is auto-cleared. After completing the Local read command, the relevant pointer value must be cleared by this bit before the next Local read command.
0	LOCAL_WR_POINTER_RST		0	No operation
			1	The LOCAL_WR_OUT_POINTER and the LOCAL_WR_TRANS_POINTER are reset. This bit is auto-cleared. After completing the Local write command, the relevant pointer value must be cleared by this bit before the next Local write command.

FAULT-AUTO-READ INTERVAL

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h11	0x0000	FAULT_AUTO_RD_INTERVAL [15:0]															

Bits		Fields	Descriptions	
15:0	FAULT_AUTO_RD_INTERVAL		This register determines the execution interval for the fault auto-read commands. Counted in MCLK units.	

FAULT-AUTO-READ EVENT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0	
'h12	0x0000	FAULT_AUTO_RD_EVENT [7:0]								x	x	x	x	x	x	x	FAULT_AUTO_RD_INTERVAL[16]	

Bits		Fields	Descriptions	
15:8	FAULT_AUTO_RD_EVENT		This register determines the execution times for the fault auto-read commands. The value is reset at VSYNC rising edge.	
0	FAULT_AUTO_RD_INTERVAL		The MSB bit of the FAULT_AUTO_RD_INTERVAL [16:0]	

INTERRUPT ENABLE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	
'h13	0x0000	INT_OV50_EN	INT_UV50_EN	INT_OV15_EN	INT_UV15_EN	INT_CMD_PC_EN	INT_ACC_CNT_ERR_EN	INT_PARITY_ERR_EN	
Addr	Default	B8		B7		B6		B5	
'h13	0x0000	INT_TIMEOUT_ERR_EN		INT_FAULT_RD_FAIL_EN		INT_FAULT_AUTO_RD_FAIL_EN		INT_RD_REC_FAIL_EN	
Addr	Default	B4		B3		B2		B1	B0
'h13	0x0000	INT_LD_EN		INT_THERMAL_EN		INT_SHORT_EN		INT_OPEN_EN	INT_FB_EN

Bits	Fields	Descriptions
15	INT_OV50_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the V_{DD} , 5V, will be go over the OV50_LEVEL value.
14	INT_UV50_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the V_{DD} , 5V, will be go under the UV50_LEVEL value.
13	INT_OV15_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the internal LDO, 1.5V, will be go over the OV50_LEVEL value.
12	INT_UV15_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the internal LDO, 1.5V, will be go under the UV50_LEVEL value.
11	INT_CMD_PC_EN	This bit enables to issue the interrupt through the CHx_PAR_ERR in case that parity error occurs while XDR communication.
10	INT_ACC_CNT_ERR_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the SPI transaction counter value on SPI data will be different from the (ACC_COUNTER[7:0] + 1)
9	INT_PARITY_ERR_EN	update
8	INT_TIMEOUT_ERR_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the timer will be expired during the execution for read-back commands. The expired time is $MCLK * 32,767$.
7	INT_FAULT_RD_FAIL_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault read-back from the XDRs.
6	INT_FAULT_AUTO_RD_FAIL_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the fault auto-read-back from the XDRs.
5	INT_RD_REC_FAIL_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that any violation happens during the execution for the local read-back from the XDRs.
4	INT_LD_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the LD_DIFF_POINTER is equal to or smaller than the LD_DIFF_THRESHOLD during the execution of the LD transfer to the XDRs.
3	INT_THERMAL_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the thermal regulation happens among the XDR drivers during dimming operation.
2	INT_SHORT_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the short condition happens among the XDR drivers during dimming operation.
1	INT_OPEN_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the open condition happens among the XDR drivers during dimming operation.
0	INT_FB_EN	This bit enables to issue the interrupt through the nINT_FAULT pin in case that the fb condition happens among the XDR drivers during dimming operation.

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COMMAND STATUS 1

Addr	Default	B15	B14	B13	B12
'h14	Read only	LOCAL_RD_DONE	LOCAL_RD_DOING	LOCAL_WR_DONE	LOCAL_WR_DOING

Addr	Default	B11	B10	B9	B8	B7	B6
'h14	Read only	GLOBAL_WR_DONE	GOLBAL_WR_DOING	LD_TRANS_DONE	LD_TRANS_DOING	FAULT_DONE	FAULT_DOING

Addr	Default	B5	B4	B3	B2	B1	B0
'h14	Read only	FAULT_AUTO_DONE	FAULT_AUTO_DOING	SYNC_AUTO_DONE	SYNC_AUTO_DOING	SYNC_DONE	SYNC_DOING

Bits	Fields	Descriptions
15	LOCAL_RD_DONE	This bit is set after the execution for the local-reading command is completed.
14	LOCAL_RD_DOING	This bit is set while the local-reading command is working until it is complete.
13	LOCAL_WR_DONE	This bit is set after the execution for the local-writing command is completed.
12	LOCAL_WR_DOING	This bit is set while the local-writing command is working until it is complete.
11	GLOBAL_WR_DONE	This bit is set after the execution for the global-writing command is completed.
10	GLOBAL_WR_DOING	This bit is set while the global-writing command is working until it is complete.
9	LD_TRANS_DONE	This bit is set after the execution for the LD transferring command is completed.
8	LD_TRANS_DOING	This bit is set while the LD transferring command is working until it is complete.
7	FAULT_DONE	This bit is set after the execution for the fault-reading command is completed.
6	FAULT_DOING	This bit is set while the fault-reading command is working until it is complete.
5	FAULT_AUTO_DONE	This bit is set after the execution for the fault auto-reading command is completed.
4	FAULT_AUTO_DOING	This bit is set while the fault auto-reading command is working until it is complete.
3	SYNC_AUTO_DONE	This bit is set after the execution for the sync auto-gen command is completed.
2	SYNC_AUTO_DOING	This bit is set while the sync auto-gen command is working until it is complete.
1	SYNC_DONE	This bit is set after the execution for the sync-gen command is completed.
0	SYNC_DOING	This bit is set while the sync-gen command is working until it is complete.

COMMAND STATUS 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h15	Read only	x	x	x	x	x	x	x	x	x	x	x	x	x	x	ID_GEN_DONE	ID_GEN_DOING

Bits	Fields	Descriptions
1	ID_GEN_DONE	This bit is set after the execution for the id-gen command is completed.
0	ID_GEN_DOING	This bit is set while the id-gen command is working until it is complete.

RECEIVE STATUS

Addr	Default	B15	B14	B13	B12
'h16	Read only	x	FAULT_REC_FAIL	FAULT_AUTO_REC_FAIL	LOCAL_REC_FAIL

Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4
'h16	Read only	x	x	x	TIMEOUT_ERR	x	x	FAULT_REC_DONE	FAULT_REC_DOING

Addr	Default	B3	B2	B1	B0
'h16	Read only	FAULT_AUTO_REC_DONE	FAULT_AUTO_REC_DOING	LOCAL_REC_DONE	LOCAL_REC_DOING

Bits	Fields	Descriptions
14	FAULT_REC_FAIL	This bit is set if any violation happens while the fault-receiving job is working.
13	FAULT_AUTO_REC_FAIL	This bit is set if any violation happens while the fault auto-receiving job is working.
12	LOCAL_REC_FAIL	This bit is set if any violation happens while the local read-receiving job is working.
8	TIMEOUT_ERR	This bit is set if the timer will be expired while the execution of read-back due to no response from the XDR devices. This bit is auto-cleared after the readout of the INTERRUPT STATUS register.
5	FAULT_REC_DONE	This bit is set after the fault-receiving job is completed.
4	FAULT_REC_DOING	This bit is set while the fault-receiving job is working until it is complete.
3	FAULT_AUTO_REC_DONE	This bit is set after the fault auto-receiving job is completed.
2	FAULT_AUTO_REC_DOING	This bit is set while the fault auto-receiving job is working until it is complete.
1	LOCAL_REC_DONE	This bit is set after the local read-receiving job is completed.
0	LOCAL_REC_DOING	This bit is set while the local read-receiving job is working until it is complete.

INTERRUPT SOURCE STATUS

Addr	Default	B15	B14	B13	B12	B11
'h17	Read only	INT_OVUV_ERR_SRC	INT_ACC_CNT_ERR_SRC	INT_PARITY_ERR_SRC	INT_TIMEOUT_ERR_SRC	INT_FAULT_REC_FAIL_SRC

Addr	Default	B10	B9	B8	B7
'h17	Read only	INT_FAULT_AUTO_REC_FAIL_SRC	INT_RD_REC_FAIL_SRC	INT_LD_TRAN_SRC	X

Addr	Default	B6	B5	B4	B3	B2	B1	B0
'h17	Read only	X	INT_SOURCE4	INT_SOURCE3	INT_SOURCE2	INT_SOURCE1	INT_LD	INT_FAULT

Bits	Fields	Descriptions
15	INT_OVUV_ERR_SRC	This bit shows whether the UV or OV detection condition of V_{DD} and internal LDO, respectively, has been happened while the XC is working. This bit is auto-cleared after the readout of INTERRUPT STATUS register.
14	INT_ACC_CNT_ERR_SRC	This bit shows whether the SPI access counter value on the SPI transaction from a Host has been different from the (ACC_COUNTER [7:0] + 1) while the SPI transactions are occurred. This bit is auto-cleared after the readout of INTERRUPT STATUS register.
13	INT_PARITY_ERR_SRC	This bit shows whether the SPI data stream's parity value on the SPI transaction from a Host has not been equal to XC internal parity status while the SPI transactions are occurred. This bit is auto-cleared after the readout of INTERRUPT STATUS register. XC internal parity is '1' when the number of '1' is odd for SPI write data.
12	INT_TIMEOUT_ERR_SRC	This bit shows whether the timeout has been happened while the read-back commands are working. This bit is auto-cleared after the readout of INTERRUPT STATUS register.
11	INT_FAULT_REC_FAIL_SRC	This bit shows whether any violation has been happened while the fault-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
10	INT_FAULT_AUTO_REC_FAIL_SRC	This bit shows whether any violation has been happened while the fault auto-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
9	INT_RD_REC_FAIL_SRC	This bit shows whether any violation has been happened while the local read-receiving job is working. This bit value will remain at the same level until start of the next command even if this register has been read-out.
8	INT_LD_TRAN_SRC	This bit shows whether the LD interrupt has been asserted while transferring the LD data to XD12R drivers through the XC24R controller. This bit value will remain at the same level until start of the next command even if this register has been read-out.

INTERRUPT SOURCE STATUS

Addr	Default	B15		B14		B13		B12		B11		
'h17	Read only	INT_OVUV_ERR_SRC		INT_ACC_CNT_ERR_SRC		INT_PARITY_ERR_SRC		INT_TIMEOUT_ERR_SRC		INT_FAULT_REC_FAIL_SRC		
Addr	Default	B10			B9			B8			B7	
'h17	Read only	INT_FAULT_AUTO_REC_FAIL_SRC			INT_RD_REC_FAIL_SRC			INT_LD_TRAN_SRC			X	
Addr	Default	B6	B5		B4		B3		B2		B1	B0
'h17	Read only	X	INT_SOURCE4		INT_SOURCE3		INT_SOURCE2		INT_SOURCE1		INT_LD	INT_FAULT

Bits	Fields	Descriptions
5	INT_SOURCE4	This bit indicate int_thermal_source from XD device for ft_mode 0 and int_fault_source for ft_mode 1,2.
4	INT_SOURCE3	This bit indicate int_shour_source from XDR device for ft_mode 0 and int_fb_source3 for ft_mode 1 and int_open_source for ft_mode2.
3	INT_SOURCE2	This bit indicate int_fb_source2 from XDR device for ft_mode 1,2 and int_open_source for ft_mode 0.
2	INT_SOURCE1	This bit indicate int_fb_source1 from XDR device for ft_mode 0~2.
1	INT_LD	If this bit is set, the nINT_LD pin will be asserted to the active-low. This bit represents an interrupt sources, INT_LD_TRAN_SRC if its interrupt enable bits is set. If the INT_LD_EN bit is set and INT_LD_TRAN_SRC is asserted, then, this bit will be asserted through nINT_LD pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out.
0	INT_FAULT	If this bit is set, the nINT_FAULT pin will be asserted to the active-low. This bit represents several interrupt sources, INT_FB_SRC, INT_OPEN_SRC, INT_SHORT_SRC, INT_TML_SRC, INT_LOCAL_REC_FAIL_SRC, INT_FAULT_AUTO_REC_FAIL_SRC, INT_FAULT_REC_FAIL_SRC, INT_TIMEOUT_ERR_SRC, INT_PARITY_ERR_SRC, INT_ACC_CNT_ERR_SRC and INT_OVUV_ERR_SRC if their own interrupt enable bits are set. If the corresponding interrupt enable bit is set and the interrupt source is asserted, then, this bit will be asserted through nINT_FAULT pin. This bit is auto-cleared after this INTERRUPT STATUS register is read-out.

CMD PARITY ERR STATUS1

Addr	Default	B15	B14	B13	B12	B11
'h18	Read only	CH16_PAR_ERR	CH15_PAR_ERR	CH14_PAR_ERR	CH13_PAR_ERR	CH12_PAR_ERR

Addr	Default	B10	B9	B8	B7
'h18	Read only	CH11_PAR_ERR	CH10_PAR_ERR	CH9_PAR_ERR	CH8_PAR_ERR

Addr	Default	B6	B5	B4	B3	B2	B1	B0
'h18	Read only	CH7_PAR_ERR	CH6_PAR_ERR	CH5_PAR_ERR	CH4_PAR_ERR	CH3_PAR_ERR	CH2_PAR_ERR	CH1_PAR_ERR

Bits	Fields	Descriptions
15	CH16_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH16 in case of INT_CMD_PC_EN is set.
14	CH15_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH15 in case of INT_CMD_PC_EN is set.
13	CH14_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH14 in case of INT_CMD_PC_EN is set.
12	CH13_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH13 in case of INT_CMD_PC_EN is set.
11	CH12_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH12 in case of INT_CMD_PC_EN is set.
10	CH11_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH11 in case of INT_CMD_PC_EN is set.
9	CH10_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH10 in case of INT_CMD_PC_EN is set.
8	CH9_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH9 in case of INT_CMD_PC_EN is set.
7	CH8_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH8 in case of INT_CMD_PC_EN is set.
6	CH7_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH7 in case of INT_CMD_PC_EN is set.
5	CH6_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH6 in case of INT_CMD_PC_EN is set.
4	CH5_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH5 in case of INT_CMD_PC_EN is set.
3	CH4_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH4 in case of INT_CMD_PC_EN is set.
2	CH3_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH3 in case of INT_CMD_PC_EN is set.
1	CH2_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH2 in case of INT_CMD_PC_EN is set.
0	CH1_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH1 in case of INT_CMD_PC_EN is set.

CMD PARITY ERR STATUS2

Addr	Default	B15	B14	B13	B12	B11
'h19	Read only	OV50_DET	UV50_DET	OV15_DET	UV15_DET	x

Addr	Default	B10	B9	B8	B7	B6
'h19	Read only	x	x	x	CH24_PAR_ERR	CH23_PAR_ERR

Addr	Default	B5	B4	B3	B2	B1	B0
'h19	Read only	CH22_PAR_ERR	CH21_PAR_ERR	CH20_PAR_ERR	CH19_PAR_ERR	CH18_PAR_ERR	CH17_PAR_ERR

Bits	Fields	Descriptions
15	OV50_DET	These bits will be set whenever the detection condition keeps for above 100us and will keep the assertion unless the detection condition is cleared.
14	UV50_DET	
13	OV15_DET	
12	UV15_DET	
7	CH24_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH24 in case of INT_CMD_PC_EN is set.
6	CH23_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH23 in case of INT_CMD_PC_EN is set.
5	CH22_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH22 in case of INT_CMD_PC_EN is set.
4	CH21_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH21 in case of INT_CMD_PC_EN is set.
3	CH20_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH20 in case of INT_CMD_PC_EN is set.
2	CH19_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH19 in case of INT_CMD_PC_EN is set.
1	CH18_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH18 in case of INT_CMD_PC_EN is set.
0	CH17_PAR_ERR	This bit will be set when parity error occurs while XDR communication on CH17 in case of INT_CMD_PC_EN is set.

SPI FAULT STATUS & CONTROL

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6
'h1A	Read only	ACC_COUNTER [7:0]								x	
Addr	Default	B5		B4		B3	B2	B1		B0	
'h1A	Read only	ACC_CNT_ERR_STATUS		PARITY_ERR_STATUS		x	x	SPI_ACC_CNT_ERR_DET_EN		SPI_PARITY_ERR_DET_EN	

Bits	Fields	Descriptions			
15:8	ACC_COUNTER	These bits show how many times the access from a Host to a XC has been executed on SPI bus. Therefore, the next access number should be same as (previous ACC_COUNTER + 1). If this counter is expired, then the counter will be roll-overed to '0' as the next start value. If the value is not same as XC internal frame counter, then the INT_ACC_CNT_ERR_SRC bit will be set under the INT_ACC_CNT_ERR_EN set. XC internal frame counter reset as '0' when power off-on and increase by 1 for any SPI commands			
5	ACC_CNT_ERR_STATUS	Read-Only. This bit shows whether the access counter value error has been happened or not. Auto-cleared after readout INTERRUPT SOURCE STATUS.			
4	PARITY_ERR_STATUS	Read-Only. This bit shows whether the parity error has been happened or not. Auto-cleared after readout INTERRUPT SOURCE STATUS.			
1	SPI_ACC_CNT_ERR_DET_EN	This bit enables the detection whether the access count value from a Host on SPI transaction is correct.			
0	SPI_PARITY_ERR_DET_EN	This bit enables the detection whether the access count value and parity from a Host on SPI transaction is correct. Extended 16bits of SPI write frame is as below.			
		<table><tr><td>SPI Data Stream</td><td>8-bit = ACC_COUNTER value</td><td>the 3rd bit of last byte = Parity bit</td></tr></table>	SPI Data Stream	8-bit = ACC_COUNTER value	the 3 rd bit of last byte = Parity bit
SPI Data Stream	8-bit = ACC_COUNTER value	the 3 rd bit of last byte = Parity bit			

CLK CONTROL 1

The XC24R features three clock sources to operate its internal digital circuits. It is highly recommended to use a single unified clock source. If individual blocks require different clocks, ensure all functions are thoroughly tested before final production. This register must be configured in the very first step.

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h1B	0x0000	X	SPREAD1_SPD [2:0]			BUF_READ_SPRD	x	SVO_CLK_SEL	OPT_CLK_SEL
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h1B	0x0000	x	SPRD1_GAIN [2:0]			SERIAL_SPRD_EN	OSC1_SPREAD_EN	SERIALIZER_SKEW_EN	x

Bits	Fields	Descriptions
14:12	SPREAD1_SPD	Refer to the Table 9.
11	BUF_READ_SPRD	update
9	SVO_CLK_SEL	'0' Operating CLK '1' Internal OSC1 Must be set as '1' if not use External MCLK
8	OPT_CLK_SEL	'0' Internal OSC2 '1' External MCLK Must be set as '1' if not use External MCLK
6:4	SPRD1_GAIN	Internal OSC clock spread period. Refer to the Table 9. '00' x1, '01' x2, '10' x4, '11' x8
3	SERIAL_SPRD_EN	'0' Operating CLK '1' Internal OSC1
2	OSC1_SPREAD_EN	Spread of internal OSC clock. (The external MCLK can not be spread) Refer to the Figure 22. for the clock scheme. '0' Not Spread '1' Spread
1	SERIALIZER_SKEW_EN	'0' Skew disabled between Serializers '1' Skew enabled

CLK CONTROL 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h1C	0x0000	x	SPREAD2_SPD [2:0]			x	x	x	x
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h1C	0x0000	x	SPRD2_GAIN [2:0]			x	OSC2_SPREAD_EN	x	reserved

Bits	Fields	Descriptions
14:12	SPREAD2_SPD	Refer to the Table 9.
6:4	SPRD2_GAIN	Internal OSC clock spread period. Refer to the Table 9. '00' x1, '01' x2, '10' x4, '11' x8
2	OSC2_SPREAD_EN	Spread of internal OSC clock. (The external MCLK can not be spread) Refer to the Figure 22. for the clock scheme. '0' Not Spread '1' Spread

SERIALIZER BAUDRATE GEN

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1D	0x0C18	x	SERIAL_CLK_LOW [6:0]								x	SERIAL_CLK_HIGH [6:0]					

Bits	Fields	Descriptions
14:8	SERIAL_CLK_LOW	These bits determine the number of MCLK of V_{DD} level in logic low value '0' or the number of MCLK of 0 level in logic high value '1' when the XC is driving the D_{IO} pin to the XDRs. For more details, refer to Table. 4, Logic low time.
6:0	SERIAL_CLK_HIGH	These bits determine the number of MCLK of V_{DD} level in logic high value '1' or the number of MCLK of 0 level in logic low value '0' when the XC is driving the D_{IO} pin to the XDRs. For more details, refer to Table. 4, Logic low time.

LATENCY

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1E	0x70C0	SERIAL_LATENCY [7:0]								CMD_LATENCY [7:0]							

Bits	Fields	Descriptions
15:8	SERIAL_LATENCY	This determines the interval time between the successive data packets incoming from the XDRs. That means the interval time between the last bit of n^{th} XDR device's data packet and the first bit of $(n+1)^{th}$ XDR device's data packet incoming from the daisied XDRs to the XC. The setting value should be larger than the 1-bit serial data length time at which the daisied XDR devices are sending to the XC. So, the proper value depends on the serial baudrate of the XDRs and can be set between (1-bit time + 3 MCLKs) and (1-bit time + 5 MCLKs). If the proper value will be set, then the serial data value will be broken. <i>It is careful that this register value should be changed to the proper value firstly in case that the Serializer Output Baudrate of XDR is changed to slow down.</i>
7:0	CMD_LATENCY	This determines the interval time between the end of the current serial data out stream and the start of a new next serial data out stream to the XDRs. <i>It is careful that this value should be at least 1.5 times larger than the SERIAL LATENCY value of XDR.</i>

TIMEOUT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h1F	0xFFFF	TIMEOUT [16:1]															

Bits	Fields	Descriptions
15:0	TIMEOUT	This register determines how many clock cycles to wait after starting read operation to XDR device.

DAISIED DEVICE CHANNEL SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h20	0x0000	x											DAISIED_DEV_CH_SIZE1 [4:0]				

Bits	Fields	Descriptions
4:0	DAISY_DEV_CH_SIZE 1	This register determines how many channels of the XDRs daisied are activated. Every XDRs daisied must have the same number of active channels but the last device in daisy can have the different number of active channels. The DAISIED_DEV_CH_SIZE1 register value will have the same effect on all other channels. This register value affects total number of LD data to each daisied XDR device. The number of LD data is multiplied value of DAISIED_DEV_CH_SIZE and LD_TYPE which is bit[2:0] of address 'h06.

DAISY SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h21	0x0000	x	DAISY_SIZE_CH3 [4:0]				DAISY_SIZE_CH2 [4:0]				DAISY_SIZE_CH1 [4:0]						

DAISY SIZE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h22	0x0000	x	DAISY_SIZE_CH6 [4:0]				DAISY_SIZE_CH5 [4:0]				DAISY_SIZE_CH4 [4:0]						

DAISY SIZE 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h23	0x0000	x	DAISY_SIZE_CH9 [4:0]				DAISY_SIZE_CH8 [4:0]				DAISY_SIZE_CH7 [4:0]						

DAISY SIZE 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h24	0x0000	x	DAISY_SIZE_CH12 [4:0]				DAISY_SIZE_CH11 [4:0]				DAISY_SIZE_CH10 [4:0]						

DAISY SIZE 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h25	0x0000	x	DAISY_SIZE_CH15 [4:0]				DAISY_SIZE_CH14 [4:0]				DAISY_SIZE_CH13 [4:0]						

DAISY SIZE 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h26	0x0000	x	DAISY_SIZE_CH18 [4:0]				DAISY_SIZE_CH17 [4:0]				DAISY_SIZE_CH16 [4:0]						

DAISY SIZE 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h27	0x0000	x	DAISY_SIZE_CH21 [4:0]				DAISY_SIZE_CH20 [4:0]				DAISY_SIZE_CH19 [4:0]						

DAISY SIZE 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h28	0x0000	x	DAISY_SIZE_CH24 [4:0]				DAISY_SIZE_CH23 [4:0]				DAISY_SIZE_CH22 [4:0]						

Bits	Fields	Descriptions
14:10 9:5 4:0	DAISY_SIZE_CHx	This register should be set with a valid number for correct operating. The number means how many XD12R drivers are daisy in the corresponding channel. The maximum 31 XD12 drivers per channel can be daisy.

BLOCK SIZE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h29	0x0000	BLOCK_SIZE_CH2 [7:0]								BLOCK_SIZE_CH1 [7:0]							

BLOCK SIZE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2A	0x0000	BLOCK_SIZE_CH4 [7:0]								BLOCK_SIZE_CH3 [7:0]							

BLOCK SIZE 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2B	0x0000	BLOCK_SIZE_CH6 [7:0]								BLOCK_SIZE_CH5 [7:0]							

BLOCK SIZE 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2C	0x0000	BLOCK_SIZE_CH8 [7:0]								BLOCK_SIZE_CH7 [7:0]							

BLOCK SIZE 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2D	0x0000	BLOCK_SIZE_CH10 [7:0]								BLOCK_SIZE_CH9 [7:0]							

BLOCK SIZE 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2E	0x0000	BLOCK_SIZE_CH12 [7:0]								BLOCK_SIZE_CH11 [7:0]							

BLOCK SIZE 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h2F	0x0000	BLOCK_SIZE_CH14 [7:0]								BLOCK_SIZE_CH13 [7:0]							

BLOCK SIZE 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h30	0x0000	BLOCK_SIZE_CH16 [7:0]								BLOCK_SIZE_CH15 [7:0]							

BLOCK SIZE 9

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h31	0x0000	BLOCK_SIZE_CH18 [7:0]								BLOCK_SIZE_CH17 [7:0]							

BLOCK SIZE 10

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h32	0x0000	BLOCK_SIZE_CH20 [7:0]								BLOCK_SIZE_CH19 [7:0]							

BLOCK SIZE 11

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h33	0x0000	BLOCK_SIZE_CH22 [7:0]								BLOCK_SIZE_CH21 [7:0]							

BLOCK SIZE 12

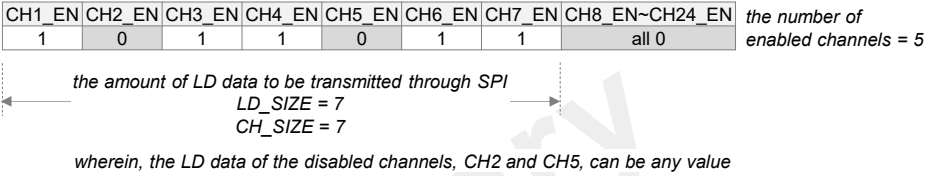
Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h34	0x0000	BLOCK_SIZE_CH24 [7:0]								BLOCK_SIZE_CH23 [7:0]							

Bits	Fields	Descriptions
15:8 7:0	BLOCK_SIZE_CHx	This register should be set with a valid number for correct operating. The number means how many block zones are daisied in the corresponding channel. The maximum 255 block zones per channel can be daisied. This value should be same as XDR daisy size multiplied by DAISIED_DEV_CH_SIZE.

CHANNEL ENABLE 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h35	0x0000	CH16_EN	CH15_EN	CH14_EN	CH13_EN	CH12_EN	CH11_EN	CH10_EN	CH9_EN
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h35	0x0000	CH8_EN	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN

Bits	Fields	Descriptions
15:0	CHx_EN	These bits should be set if the corresponding channels are to be used, but can not be set if any channel is not used. The number of enabled channels in these registers does not need to match the LD_SIZE value. It is careful that the number of enabled channels should be the same as the LD_SIZE and CH_SIZE value, but the number can be smaller than the LD_SIZE value like the special mapping below. If Channel size is equal to number of channel enable, LD data from SPI can be omitted for disable channel. And it is recommended not to omit Channel 1, in other words CH1_EN register should be set as '1'.



CHANNEL ENABLE 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h36	0x0000	LD_WIDTH [1:0]		x	CH_SIZE [4:0]				
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h36	0x0000	CH24_EN	CH23_EN	CH22_EN	CH21_EN	CH20_EN	CH19_EN	CH18_EN	CH17_EN

Bits	Fields	Descriptions
15:14	LD_WIDTH	These bits represent the number of bits contained in one LD data in one block zone. 2'b00 : NOP 2'b01 : 12 bits/block 2'b10 : 14 bits/block 2'b11 : 16 bits/block
12:8	CH_SIZE	This value determines the amount of LD data size of one line to be transmitted through the SPI interface. This value is not needed to match with the number of enabled channels in the CHANNEL ENABLE register, but it keeps in mind that the LD data to be transmitted will be mapped sequentially from channel 1 to channel n regardless of the corresponded CHx_EN bit activation.
7:0	CHx_EN	The same as above CHANNEL ENABLE 1 register.

FLLCNT1-1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h37	0x0000	FLL1_CNT [15:0]															

Bits	Fields	Descriptions
15:0	FLL1_CNT [15:0]	This determines the FLL1 target frequency value. OSC1 target frequency[Hz] = Vsync[Hz] * FLLCNT[20:0] OSC1 is for the serialize output.

FLLCNT1-2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h38	0x0000	FLL1_EN	FLL1_SYNC	FLL1_RANGE[1:0]		x	x	FLL1_ERR_RANGE[1:0]	

Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h38	0x0000	x	x	x	FLL1_CNT [20:16]				

Bits	Fields	Descriptions
15	FLL1_EN	This bit enables the FLL1 function. The FLL1 continues to calibrate the MCLK frequency to exactly match the PWM frequency with the time of one frame. 1 Enable FLL1 function (= The MCLK is unlocking, so calibrated to the target frequency) 0 Disable FLL function (= The MCLK is locking at the current frequency) <i>It is needed that this bit should be set to 0 just before the first vsync starts to be entered to the XC24R and set to 1 just before the vsync is not asserted if the vsync will be not supplied any more to the XC24R .</i>
14	FLL1_SYNC	Locking sync pad input selection '0' VSYNCl (recommended) '1' FLLSYNC (Async)
13:12	FLL1_RANGE	It is recommended to use the default value.
9:8	FLL1_ERR_RANGE	Error clamping range between measured counting and FLL1_CNT within fill sync duration.
4:0	FLL1_CNT [20:16]	This determines the FLL1 target frequency value.

FLLCNT2-1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h39	0x0000	FLL2_CNT [15:0]															

Bits	Fields	Descriptions
15:0	FLL2_CNT [15:0]	This determines the OSC2 target frequency value. OSC2 is for the main operating clk.

FLLCNT2-2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h3A	0x0000	FLL2_EN	FLL2_SYNC	FLL2_RANGE[1:0]		x	x	FLL2_ERR_RANGE[1:0]	
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h3A	0x0000	x	x	x	FLL2_CNT [20:16]				

Bits	Fields	Descriptions
15	FLL2_EN	This bit enables the FLL2 function. The FLL2 continues to calibrate the MCLK frequency to exactly match the PWM frequency with the time of one frame. 1 Enable FLL2 function (= The MCLK is unlocking, so calibrated to the target frequency) 0 Disable FLL function (= The MCLK is locking at the current frequency) <i>It is needed that this bit should be set to 0 just before the first vsync starts to be entered to the XC24R and set to 1 just before the vsync is not asserted if the vsync will be not supplied any more to the XC24R .</i>
14	FLL2_SYNC	Locking sync pad input selection '0' VSYNCl (recommended) '1' FLLSYNC (Async)
13:12	FLL2_RANGE	It is recommended to use the default value.
9:8	FLL2_ERR_RANGE	Error clamping range between measured counting and FLL1_CNT within fill sync duration.
4:0	FLL2_CNT [20:16]	This determines the FLL2 target frequency value.

VO DELAY

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h3B	0x7000	x	GATE3_POL	GATE2_POL	GATE1_POL	x	x	x	x
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h3B	0x7000	SVO_DELAY							

Bits	Fields	Descriptions
14	GATE3_POL	'1' positive polarity '0' negative polarity
13	GATE2_POL	'1' positive polarity '0' negative polarity
12	GATE1_POL	'1' positive polarity '0' negative polarity
7:0	SVO_DELAY	update

VO OFF/ON

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3C	0x0000	VO_OFF [7:0]								VO_ON [7:0]							

Bits	Fields	Descriptions
15:8	VO_OFF	Vsync pulse off time in front of sub vsync output from SVO
7:0	VO_ON	Vsync pulse on time in front of sub vsync output from SVO

GATE CONTROL

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFA	0x0000	X									GATE3H_EN	GATE2H_EN	GATE1H_EN	X	GATE3L_EN	GATE2L_EN	GATE1L_EN

Bits	Fields	Descriptions
6	GATE3H_EN	'1' : activate GATE3H signal, '0' : Deactivate GATE3H signal (Output is tied to GND)
5	GATE2H_EN	'1' : activate GATE2H signal, '0' : Deactivate GATE2H signal (Output is tied to GND)
4	GATE1H_EN	'1' : activate GATE1H signal, '0' : Deactivate GATE1H signal (Output is tied to GND)
2	GATE3L_EN	'1' : activate GATE3L signal, '0' : Deactivate GATE3L signal (Output is tied to GND)
1	GATE2L_EN	'1' : activate GATE2L signal, '0' : Deactivate GATE2L signal (Output is tied to GND)
0	GATE1L_EN	'1' : activate GATE1L signal, '0' : Deactivate GATE1L signal (Output is tied to GND)

GATE signals are for controlling VLED power on/off switch,
After vsync comes into XC24R,
then GATE signal stats to generate as many as SVO_NO('h41)

GATE1 OFFSET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFB	0x0000	GATE1_OFFSET2 [7:0]								GATE1_OFFSET1 [7:0]							

Bits	Fields	Descriptions
15:8	GATE1_OFFSET2	Offset time from GATE1H to GATE1L
7:0	GATE1_OFFSET1	Offset time from GATE1L to GATE1H, , GATE1H is corresponding to H_G1 pad and GATE1L is corresponding to L_G1. Other gate related signals are the same naming rule. GATExL signal is for discharge of LED power source, this offset value is for preventing from short path between VLED sourcing path and discharging path. GATE1_OFFSET1 is interval time between GATE1 falling to GATE1_L rising. The unit of setting value is clk.

GATE2 OFFSET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFC	0x0000	GATE2_OFFSET2 [7:0]								GATE2_OFFSET1 [7:0]							

Bits	Fields	Descriptions
15:8	GATE2_OFFSET2	Offset time from GATE2H to GATE2L
7:0	GATE2_OFFSET1	Offset time from GATE2L to GATE2H

GATE3 OFFSET

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFD	0x0000	GATE3_OFFSET2 [7:0]								GATE3_OFFSET1 [7:0]							

Bits	Fields	Descriptions
15:8	GATE3_OFFSET2	Offset time from GATE2H to GATE2L
7:0	GATE3_OFFSET1	Offset time from GATE2L to GATE2H

SV VAR CONTROL1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFE	0x0000	SV_VAR_SIZE1 [7:0]								X							

Bits	Fields	Descriptions
15:8	SV_VAR_SIZE1	Random variation size masking bit of GATE1 and related SVO

SV VAR CONTROL2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hFF	0x0000	SV_VAR_SIZE3 [7:0]								SV_VAR_SIZE2 [7:0]							

Bits	Fields	Descriptions
15:8	SV_VAR_SIZE3	Random variation size masking bit of GATE3 and related SVO
7:0	SV_VAR_SIZE2	Random variation size masking bit of GATE2 and related SVO

SVO ON

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3D	0x0000	x	x	x	x	SVO_ON [11:0]											

Bits	Fields	Descriptions
11:0	SVO_ON	Sub vsync high pulse width duration. The unit of register value is clk.

SVO1 OFF

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3E	0x0000	SVO1_OFF [15:0]															

Bits	Fields	Descriptions
15:0	SVO1_OFF	Sub vsync low pulse width duration related GATE1 signal. The unit of register value is clk.

SVO2 OFF

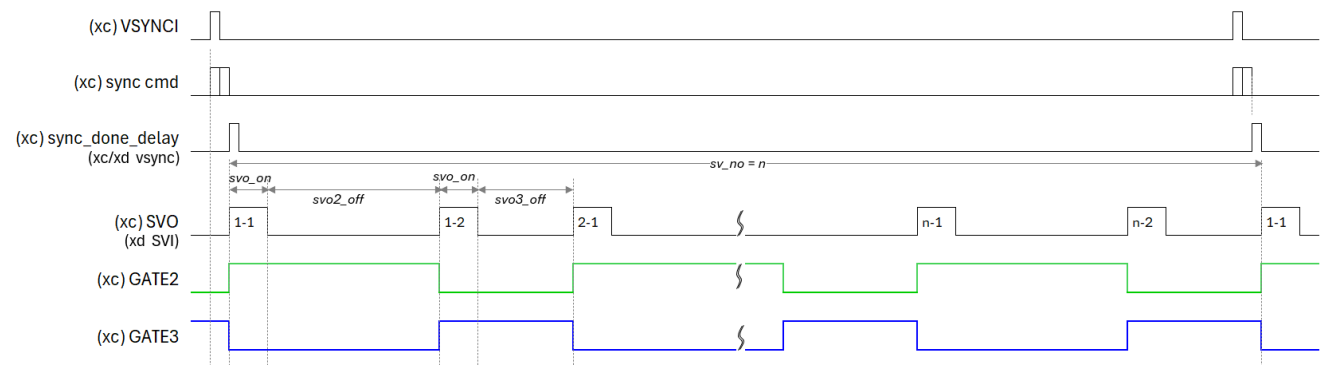
Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h3F	0x0000	SVO2_OFF [15:0]															

Bits	Fields	Descriptions
15:0	SVO2_OFF	Sub vsync off duration related GATE2 signal

SVO3 OFF

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h40	0x0000	SVO3_OFF [15:0]															

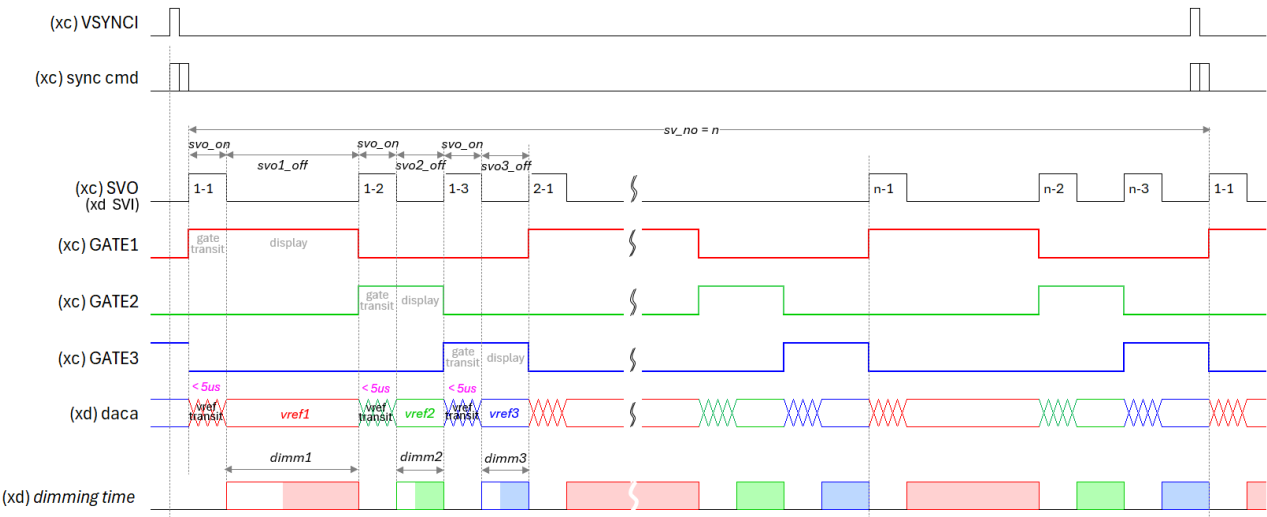
Bits	Fields	Descriptions
15:0	SVO3_OFF	Sub vsync off duration related GATE3 signal, please refer to following timing chart.



SVO NUMBER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h41	0x0000	x	x	x	x	x	x	SV_NO_MULT [1:0]		x	x	SV_NO [5:0]					

Bits	Fields	Descriptions
9:8	SV_NO_MULT	'd2 : svo_off 2/3 active 'd3 : svo_off 1/2/3 active
5:0	SV_NO	Sub vsync repeat number Total svo number is SV_NO multiplied by SV_NO_TYPE



DAC NF CONTROL

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h42	0x0000	x	x	DAC_LVL[1:0]		x	x	x	x	x	x	x	x	x	x	X	DGRJT_EN

Bits	Fields	Descriptions
13:12	DAC_LVL	DAC level for FB control output
0	DGRJT_EN	'0' data glitch rejection disable '1' data glitch rejection enable

DAC CONTROL

Addr	Default	B15	B14	B13	B12	B11	B10
'h43	0x0707	DAC_THERMAL_OFF	DAC3_HOLD_EN	DAC2_HOLD_EN	DAC1_HOLD_EN	x	DAC3_DEC1_MODE

Addr	Default	B9	B8	B7	B6	B5	B4
'h43	0x0707	DAC2_DEC1_MODE	DAC1_DEC1_MODE	DAC3_FB_MODE	DAC2_FB_MODE	DAC1_FB_MODE	DAC_SYNC_MODE

Addr	Default	B3	B2	B1	B0
'h43	0x0707	DAC_AUTO_TYPE	DAC3_AUTO	DAC2_AUTO	DAC1_AUTO

Bits	Fields	Descriptions
15	DAC_THERMAL_OFF	This bit enables the DAC OUT to the initial value if the thermal event from XDRs is occurred.
14	DAC3_HOLD_EN	This bit enables to fix the DAC OUT to the current value. This can be activated in case that (CURR_TGT_DAC - current DAC OUT) value successively stays below DAC_INC_HOLD_THRESHOLD for DAC_INC_HOLD_LIMIT number which is counted per frame. This function makes the DAC OUT stay at current DAC value even though the dimming data is all zero.
13	DAC2_HOLD_EN	0 The DAC OUT is decreased by the DAC_DEC1 whenever the FB is asserted. 1 The DAC OUT is decreased by the arithmetic exponential value of DAC_DEC1 whenever the FB from the XDRs is asserted.
12	DAC1_HOLD_EN	0 The DAC OUT change results from the internal FB status. 1 The DAC OUT change results from the external FBI pin status.
10	DAC3_DEC1_MODE	0 The DAC OUT is synchronized when the SYNC commands are completed to send out to the XDRs. 1 The DAC OUT is synchronized at the VSYNCl input of the XC.
9	DAC2_DEC1_MODE	0 The DAC OUT is decreased by the DAC_DEC1 whenever the FB is asserted. 1 The DAC OUT is decreased by the arithmetic exponential value of DAC_DEC1 whenever the FB from the XDRs is asserted.
8	DAC1_DEC1_MODE	0 The DAC OUT is decreased by the DAC_DEC1 whenever the FB is asserted. 1 The DAC OUT is decreased by the arithmetic exponential value of DAC_DEC1 whenever the FB from the XDRs is asserted.
7	DAC3_FB_MODE	0 The DAC OUT change results from the internal FB status. 1 The DAC OUT change results from the external FBI pin status.
6	DAC2_FB_MODE	0 The DAC OUT change results from the internal FB status. 1 The DAC OUT change results from the external FBI pin status.
5	DAC1_FB_MODE	0 The DAC OUT change results from the internal FB status. 1 The DAC OUT change results from the external FBI pin status.
4	DAC_SYNC_MODE	0 The DAC OUT is synchronized when the SYNC commands are completed to send out to the XDRs. 1 The DAC OUT is synchronized at the VSYNCl input of the XC.

DAC CONTROL

Addr	Default	B15	B14	B13	B12	B11	B10
'h43	0x0707	DAC_FAULT_OFF	DAC3_HOLD_EN	DAC2_HOLD_EN	DAC1_HOLD_EN	x	DAC3_DEC1_MODE

Addr	Default	B9	B8	B7	B6	B5	B4
'h43	0x0707	DAC2_DEC1_MODE	DAC1_DEC1_MODE	DAC3_FB_MODE	DAC2_FB_MODE	DAC1_FB_MODE	DAC_SYNC_MODE

Addr	Default	B3	B2	B1	B0
'h43	0x0707	DAC_AUTO_TYPE	DAC3_AUTO	DAC2_AUTO	DAC1_AUTO

Bits	Fields		Descriptions
3	DAC_AUTO_TYPE	0	The DAC output fit with multiple step and apply increment step as DAC_INC3.
		1	The DAC output fit with simple step and apply increment step as DAC_INC4.
2	DAC3_AUTO	0	The DAC OUT outputs the same value as the CURR_TGT_DAC at frame sync.
		1	The DAC OUT outputs the automatically optimized value according to the FB status. The DAC OUT will be calculated every frame along with the relevant registers' set value.
1	DAC2_AUTO	0	The DAC OUT outputs the same value as the CURR_TGT_DAC at frame sync.
		1	The DAC OUT outputs the automatically optimized value according to the FB status. The DAC OUT will be calculated every frame along with the relevant registers' set value.
0	DAC1_AUTO	0	The DAC OUT outputs the same value as the CURR_TGT_DAC at frame sync.
		1	The DAC OUT outputs the automatically optimized value according to the FB status. The DAC OUT will be calculated every frame along with the relevant registers' set value.

CURRENT TARGET DAC1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h44	0x0FA0	x	x	x	x	CURR_TGT_DAC1 [11:0]											

Bits	Fields	Descriptions
11:0	CURR_TGT_DAC1	This register determines the target DAC value of the corresponding frame. This value is effective from a next frame sync, so this value should be set before a next frame sync. The written value can be visible after a next frame sync is occurred. The target value can be calculated from (LED V_F + min V_{HD} of the XDR). The initial value at the soft reset, RST1 bit=1, is the same as the DAC_MAX_LIMIT.

CURRENT TARGET DAC2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h45	0x0FA0	x	x	x	x	CURR_TGT_DAC2 [11:0]											

Bits	Fields	Descriptions
11:0	CURR_TGT_DAC2	This register determines the target DAC value of the corresponding frame. This value is effective from a next frame sync, so this value should be set before a next frame sync. The written value can be visible after a next frame sync is occurred. The target value can be calculated from (LED V_F + min V_{HD} of the XDR). The initial value at the soft reset, RST1 bit=1, is the same as the DAC_MAX_LIMIT.

CURRENT TARGET DAC3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h46	0x0FA0	x	x	x	x	CURR_TGT_DAC3 [11:0]											

Bits	Fields	Descriptions
11:0	CURR_TGT_DAC3	This register determines the target DAC value of the corresponding frame. This value is effective from a next frame sync, so this value should be set before a next frame sync. The written value can be visible after a next frame sync is occurred. The target value can be calculated from (LED V_F + min V_{HD} of the XDR). The initial value at the soft reset, RST1 bit=1, is the same as the DAC_MAX_LIMIT.

PREVIOUS TARGET DAC1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h47	0x0FA0	x	x	x	x	PREV_TGT_DAC1 [11:0]											

Bits	Fields	Descriptions
11:0	PREV_TGT_DAC1	This register shows the target DAC value of the previous frame. This value is also valid just after a next frame sync is asserted. The initial value at the soft reset, RST1 bit=1, is the DAC_MAX_LIMIT.

PREVIOUS TARGET DAC2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h48	0x0FA0	x	x	x	x	PREV_TGT_DAC2 [11:0]											

Bits	Fields	Descriptions
11:0	PREV_TGT_DAC2	This register shows the target DAC value of the previous frame. This value is also valid just after a next frame sync is asserted. The initial value at the soft reset, RST1 bit=1, is the DAC_MAX_LIMIT.

PREVIOUS TARGET DAC3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h49	0x0FA0	x	x	x	x	PREV_TGT_DAC3 [11:0]											

Bits	Fields	Descriptions
11:0	PREV_TGT_DAC3	This register shows the target DAC value of the previous frame. This value is also valid just after a next frame sync is asserted. The initial value at the soft reset, RST1 bit=1, is the DAC_MAX_LIMIT.

DAC1 OUT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4A	0x0FA0	x	x	x	x	DAC1_OUT [11:0]											

Bits	Fields	Descriptions
11:0	DAC1_OUT	This value shows the current DAC out value. The initial value after POR is the DAC_MAX_LIMIT.

DAC2 OUT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4B	0x0FA0	x	x	x	x	DAC2_OUT [11:0]											

Bits	Fields	Descriptions
11:0	DAC2_OUT	This value shows the current DAC out value. The initial value after POR is the DAC_MAX_LIMIT.

DAC3 OUT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4C	0x0FA0	x	x	x	x	DAC3_OUT [11:0]											

Bits	Fields	Descriptions
11:0	DAC3_OUT	This value shows the current DAC out value. The initial value after POR is the DAC_MAX_LIMIT.

DAC1 STATE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4D	0x0000	x	x	x	x	x	x	x	x	DAC1_STATE [3:0]				DAC1_ST_EN [3:0]			

Bits	Fields	Descriptions
7:4	DAC1_STATE	This is for test only.
3:0	DAC1_ST_EN	This is for test only.

DAC2 STATE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4E	0x0000	x	x	x	x	x	x	x	x	DAC2_STATE [3:0]				DAC2_ST_EN [3:0]			

Bits	Fields	Descriptions
7:4	DAC2_STATE	This is for test only.
3:0	DAC2_ST_EN	This is for test only.

DAC3 STATE

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h4F	0x0000	x	x	x	x	x	x	x	x	DAC3_STATE [3:0]				DAC3_ST_EN [3:0]			

Bits	Fields	Descriptions
7:4	DAC3_STATE	This is for test only.
3:0	DAC3_ST_EN	This is for test only.

DAC INCREMENT 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h50	0x40C2	DAC1_INC3 [5:0]						DAC1_INC2 [4:0]						DAC1_INC1 [4:0]			

Bits	Fields	Descriptions
15:10	DAC1_INC3	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(CURR_TGT_DAC - SOA_P3) \leq$ the current DAC OUT < $(CURR_TGT_DAC - SOA_P2)$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
9:5	DAC1_INC2	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(CURR_TGT_DAC - SOA_P2) \leq$ the current DAC OUT < $(CURR_TGT_DAC - SOA_P1)$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
4:0	DAC1_INC1	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, the current DAC OUT $\geq (CURR_TGT_DAC - SOA_P1)$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC1 INCREMENT 2 & HOLD_LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h51	0x0820	DAC1_INC_HOLD_LIMIT [8:0]										DAC_INC4 [6:0]					

Bits	Fields	Descriptions
15:7	DAC1_INC_HOLD_LIMIT	This value determines the number of frames without the FB assertion in which the current DAC stays within the range, (CURR_TGR_DAC – the current DAC) < DAC_INC_HOLD_THRESHOLD. If the counted value is same as this value, then the DAC_OUT will be held at the current DAC value so that the DAC_OUT will be not increased any more. This function is useful in case that the PWM data are kept to 0 for a lot of frames.
6:0	DAC_INC4	This value determines the amount of the DAC_OUT increasement in case that the current DAC_OUT stays within the range, the current DAC_OUT < (CURR_TGT_DAC – SOA_P3) without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC1 DECREMENT & INC WAIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h52	0x0400	x	DAC1_DEC1 [6:0]								x	DAC1_INC1_WAIT [6:0]					

Bits	Fields	Descriptions
14:8	DAC1_DEC1	This value determines the amount of the DAC_OUT decrement whenever the FB is asserted. The DAC_DEC1_MODE bit allows two modes choice, one is to decrease by the amount of DAC_DEC1, and the other is to decrease by the amount of arithmetic exponential of DAC_DEC1. Refer to the Figure 28.
6:0	DAC1_INC1_WAIT	This value determines the number of frames that the DAC_OUT increment execution will wait with the FB deassertion. In case the value is set to 2, then the DAC_OUT will be increased by the DAC_INC1 at the third frame if three successive frames are kept without the FB assertion. If the value is set to 0, then the DAC_OUT will be increased immediately whenever the frame is without a FB assertion.

DAC1 INCREMENT HOLD THRESHOLD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h53	0x020A	x	x	x	x	x	x	SIGNED	DAC1_INC_HOLD_THRESHOLD [8:0]								

Bits	Fields	Descriptions
9	SIGNED	The signed bit of the DAC_INC_HOLD_THRESHOLD. The negative value can be set to '1'.
8:0	DAC1_INC_HOLD_THRESHOLD	This determines the range that the DAC_INC_HOLD_LIMIT counter will work. Refer to the DAC_INC_HOLD_LIMIT and the DAC_HOLD_EN bit in details.

DAC1 FB VALID TIMER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h54	0x0000	DAC1_FB_VAILD_TIMER [17:2]															

Bits	Fields	Descriptions
15:0	DAC1_FB_VAILD_TIMER	This determines when the FB status is valid. This is the interval time from the frame sync to the FB valid. The setting value in this register is counted as much as (the value x MCLK x 4).

DAC1 MIN LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h55	0x0190	x	x	x	x	DAC1_MIN_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC1_MIN_LIMIT	This determines the minimum value of the DAC OUT. <i>It is recommended to set the value to decimal 100 or higher.</i>

DAC1 MAX LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h56	0x0FA0	x	x	x	x	DAC1_MAX_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC1_MAX_LIMIT	This determines the maximum value of the DAC OUT. <i>It is recommended to set the value to decimal 4,000 or lower.</i>

DAC2 INCREMENT 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h57	0x40C2	DAC2_INC3 [5:0]						DAC2_INC2 [4:0]						DAC2_INC1 [4:0]			

Bits	Fields	Descriptions
15:10	DAC2_INC3	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(\text{CURR_TGT_DAC} - \text{SOA_P3}) \leq$ the current $\text{DAC OUT} < (\text{CURR_TGT_DAC} - \text{SOA_P2})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
9:5	DAC2_INC2	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(\text{CURR_TGT_DAC} - \text{SOA_P2}) \leq$ the current $\text{DAC OUT} < (\text{CURR_TGT_DAC} - \text{SOA_P1})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
4:0	DAC2_INC1	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, the current $\text{DAC OUT} \geq (\text{CURR_TGT_DAC} - \text{SOA_P1})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC2 INCREMENT 2 & HOLD_LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h58	0x0820	DAC2_INC_HOLD_LIMIT [8:0]								DAC2_INC4 [6:0]							

Bits	Fields	Descriptions
15:7	DAC2_INC_HOLD_LIMIT	This value determines the number of frames without the FB assertion in which the current DAC stays within the range, (CURR_TGR_DAC – the current DAC) < DAC_INC_HOLD_THRESHOLD. If the counted value is same as this value, then the DAC_OUT will be held at the current DAC value so that the DAC_OUT will be not increased any more. This function is useful in case that the PWM data are kept to 0 for a lot of frames.
6:0	DAC2_INC4	This value determines the amount of the DAC_OUT increasement in case that the current DAC_OUT stays within the range, the current DAC_OUT < (CURR_TGT_DAC – SOA_P3) without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC2 DECREMENT & INC WAIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h59	0x0400	x	DAC2_DEC1 [6:0]								x	DAC2_INC1_WAIT [6:0]					

Bits	Fields	Descriptions
14:8	DAC2_DEC1	This value determines the amount of the DAC_OUT decrement whenever the FB is asserted. The DAC_DEC1_MODE bit allows two modes choice, one is to decrease by the amount of DAC_DEC1, and the other is to decrease by the amount of arithmetic exponential of DAC_DEC1. Refer to the Figure 28.
6:0	DAC2_INC1_WAIT	This value determines the number of frames that the DAC_OUT increment execution will wait with the FB deassertion. In case the value is set to 2, then the DAC_OUT will be increased by the DAC_INC1 at the third frame if three successive frames are kept without the FB assertion. If the value is set to 0, then the DAC_OUT will be increased immediately whenever the frame is without a FB assertion.

DAC2 INCREMENT HOLD THRESHOLD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5A	0x020A	x	x	x	x	x	x	SIGNED	DAC2_INC_HOLD_THRESHOLD [8:0]								

Bits	Fields	Descriptions
9	SIGNED	The signed bit of the DAC_INC_HOLD_THRESHOLD. The negative value can be set to '1'.
8:0	DAC2_INC_HOLD_THRESHOLD	This determines the range that the DAC_INC_HOLD_LIMIT counter will work. Refer to the DAC_INC_HOLD_LIMIT and the DAC_HOLD_EN bit in details.

DAC2 FB VALID TIMER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5B	0x0000	DAC2_FB_VAILD_TIMER [17:2]															

Bits	Fields	Descriptions
15:0	DAC2_FB_VAILD_TIMER	This determines when the FB status is valid. This is the interval time from the frame sync to the FB valid. The setting value in this register is counted as much as (the value x MCLK x 4).

DAC2 MIN LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5C	0x0190	x	x	X	x	DAC2_MIN_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC2_MIN_LIMIT	This determines the minimum value of the DAC OUT. <i>It is recommended to set the value to decimal 100 or higher.</i>

DAC2 MAX LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5D	0x0FA0	x	x	X	x	DAC2_MAX_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC2_MAX_LIMIT	This determines the maximum value of the DAC OUT. <i>It is recommended to set the value to decimal 4,000 or lower.</i>

DAC3 INCREMENT 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5E	0x40C2	DAC3_INC3 [5:0]						DAC3_INC2 [4:0]						DAC3_INC1 [4:0]			

Bits	Fields	Descriptions
15:10	DAC3_INC3	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(\text{CURR_TGT_DAC} - \text{SOA_P3}) \leq$ the current DAC OUT < $(\text{CURR_TGT_DAC} - \text{SOA_P2})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
9:5	DAC3_INC2	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, $(\text{CURR_TGT_DAC} - \text{SOA_P2}) \leq$ the current DAC OUT < $(\text{CURR_TGT_DAC} - \text{SOA_P1})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.
4:0	DAC3_INC1	This value determines the amount of the DAC OUT increasement in case that the current DAC OUT stays within the range, the current DAC OUT $\geq (\text{CURR_TGT_DAC} - \text{SOA_P1})$ without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC3 INCREMENT 2 & HOLD_LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h5F	0x0820	DAC3_INC_HOLD_LIMIT [8:0]										DAC3_INC4 [6:0]					

Bits	Fields	Descriptions
15:7	DAC3_INC_HOLD_LIMIT	This value determines the number of frames without the FB assertion in which the current DAC stays within the range, (CURR_TGR_DAC – the current DAC) < DAC_INC_HOLD_THRESHOLD. If the counted value is same as this value, then the DAC_OUT will be held at the current DAC value so that the DAC_OUT will be not increased any more. This function is useful in case that the PWM data are kept to 0 for a lot of frames.
6:0	DAC3_INC4	This value determines the amount of the DAC_OUT increasement in case that the current DAC_OUT stays within the range, the current DAC_OUT < (CURR_TGT_DAC – SOA_P3) without any FB assertion. The increment value is added whenever the FB is not asserted every frame. Refer to the Figure 28.

DAC3 DECREMENT & INC WAIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h60	0x0400	x	DAC3_DEC1 [6:0]								x	DAC3_INC1_WAIT [6:0]					

Bits	Fields	Descriptions
14:8	DAC3_DEC1	This value determines the amount of the DAC_OUT decrement whenever the FB is asserted. The DAC_DEC1_MODE bit allows two modes choice, one is to decrease by the amount of DAC_DEC1, and the other is to decrease by the amount of arithmetic exponential of DAC_DEC1. Refer to the Figure 28.
6:0	DAC3_INC1_WAIT	This value determines the number of frames that the DAC_OUT increment execution will wait with the FB deassertion. In case the value is set to 2, then the DAC_OUT will be increased by the DAC_INC1 at the third frame if three successive frames are kept without the FB assertion. If the value is set to 0, then the DAC_OUT will be increased immediately whenever the frame is without a FB assertion.

DAC3 INCREMENT HOLD THRESHOLD

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h61	0x020A	x	x	x	x	x	x	SIGNED	DAC3_INC_HOLD_THRESHOLD [8:0]								

Bits	Fields	Descriptions
9	SIGNED	The signed bit of the DAC_INC_HOLD_THRESHOLD. The negative value can be set to '1'.
8:0	DAC3_INC_HOLD_THRESHOLD	This determines the range that the DAC_INC_HOLD_LIMIT counter will work. Refer to the DAC_INC_HOLD_LIMIT and the DAC_HOLD_EN bit in details.

DAC3 FB VALID TIMER

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h62	0x0000	DAC3_FB_VAILD_TIMER [17:2]															

Bits	Fields	Descriptions
15:0	DAC3_FB_VAILD_TIMER	This determines when the FB status is valid. This is the interval time from the frame sync to the FB valid. The setting value in this register is counted as much as (the value x MCLK x 4).

DAC3 MIN LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h63	0x0190	x	x	X	x	DAC3_MIN_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC3_MIN_LIMIT	This determines the minimum value of the DAC OUT. <i>It is recommended to set the value to decimal 100 or higher.</i>

DAC3 MAX LIMIT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h64	0x0FA0	x	x	X	x	DAC3_MAX_LIMIT [11:0]											

Bits	Fields	Descriptions
11:0	DAC3_MAX_LIMIT	This determines the maximum value of the DAC OUT. <i>It is recommended to set the value to decimal 4,000 or lower.</i>

OSC FLL MAN_A1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h65	0x8010	OSC_MAN_EN_A	x	x	x	x	x	x	DAC_RNG_A
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h65	0x8010	x	x	FLT_CTL_A [1:0]		x	x	FLT_GAIN_A [1:0]	

Bits	Fields	Descriptions
15	OSC_MAN_EN_A	This determines the FLL_A target frequency value in manual mode.
8	DAC_RNG_A	It is recommended to use the default value.
5:4	FLT_CTL_A	It is recommended to set to 1.
1:0	FLT_GAIN_A	It is recommended to use the default value.

OSC FLL MAN_A2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h66	0x8000	OSC_FLL_MAN_A [15:0]															

Bits	Fields	Descriptions
15:0	OSC_FLL_MAN_A	This determines the FLL_A target frequency value in manual mode.

OSC FLL MAN_B1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8
'h67	0x8010	OSC_MAN_EN_B	x	x	x	x	x	x	DAC_RNG_B
Addr	Default	B7	B6	B5	B4	B3	B2	B1	B0
'h67	0x8010	x	x	FLT_CTL_B [1:0]		x	x	FLT_GAIN_B [1:0]	

Bits	Fields	Descriptions
15	OSC_MAN_EN_B	This determines the FLL_B target frequency value in manual mode.
8	DAC_RNG_B	It is recommended to use the default value.
5:4	FLT_CTL_B	It is recommended to set to 1.
1:0	FLT_GAIN_B	It is recommended to use the default value.

OSC FLL MAN_B2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h68	0x8000	OSC_FLL_MAN_B [15:0]															

Bits	Fields	Descriptions
15:0	OSC_FLL_MAN_B	This determines the FLL_B target frequency value in manual mode.

GLOBAL WRITE DATA

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h69	0x0000	x	x	x	x	DATA [11:0]											

Bits	Fields	Descriptions
11:0	GLOBAL_WR_DATA	This register represents the data to be written to the XD12 drivers daisied. This register should be written with desired value before the global-writing command is executed.

GLOBAL FAULT-READ DATA 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6A	0x0000	FAULT_CH4 [3:0]				FAULT_CH3 [3:0]				FAULT_CH2 [3:0]				FAULT_CH1 [3:0]			

GLOBAL FAULT-READ DATA 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6B	0x0000	FAULT_CH8 [3:0]				FAULT_CH7 [3:0]				FAULT_CH6 [3:0]				FAULT_CH5 [3:0]			

GLOBAL FAULT-READ DATA 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6C	0x0000	FAULT_CH12 [3:0]				FAULT_CH11 [3:0]				FAULT_CH10 [3:0]				FAULT_CH9 [3:0]			

GLOBAL FAULT-READ DATA 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6D	0x0000	FAULT_CH16 [3:0]				FAULT_CH15 [3:0]				FAULT_CH14 [3:0]				FAULT_CH13 [3:0]			

GLOBAL FAULT-READ DATA 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6E	0x0000	FAULT_CH20 [3:0]				FAULT_CH19 [3:0]				FAULT_CH18 [3:0]				FAULT_CH17 [3:0]			

GLOBAL FAULT-READ DATA 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h6F	0x0000	FAULT_CH24 [3:0]				FAULT_CH23 [3:0]				FAULT_CH22 [3:0]				FAULT_CH21 [3:0]			

Bits	Fields	Descriptions
15:12 11:8 7:4 3:0	FAULT_CHx	These bits represent the fault status from all XD12R drivers daisied in each channel. FAULT_CHx [3] : The THERMAL status, all bits from each channel are OR-ed. FAULT_CHx [2] : The SHORT status, all bits from each channel are OR-ed. FAULT_CHx [1] : The OPEN status, all bits from each channel are OR-ed. FAULT_CHx [0] : The FB status, all bits from each channel are OR-ed.

The value will be kept the current state until the value is newly updated from a next FAULT-(AUTO)-READ execution. The OR-ed signal of the FAULT_CHx [0] is driven directly to the FB pin and the pattern depends on the FB_PWM_EN bit.

PORT1 LOCAL-RW DATA 1 ~ PORT8 LOCAL RW DATA 1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h70~'h77	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 1 [11:0] ~ PORT8_LOCAL_RW_DATA 1 [11:0]											

PORT1 LOCAL-RW DATA 2 ~ PORT8 LOCAL RW DATA 2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h78~'h7F	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 2 [11:0] ~ PORT8_LOCAL_RW_DATA 2 [11:0]											

PORT1 LOCAL-RW DATA 3 ~ PORT8 LOCAL RW DATA 3

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h80~'h87	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 3 [11:0] ~ PORT8_LOCAL_RW_DATA 3 [11:0]											

PORT1 LOCAL-RW DATA 4 ~ PORT8 LOCAL RW DATA 4

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h88~'h8F	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 4 [11:0] ~ PORT8_LOCAL_RW_DATA 4 [11:0]											

PORT1 LOCAL-RW DATA 5 ~ PORT8 LOCAL RW DATA 5

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h90~'h97	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 5 [11:0] ~ PORT8_LOCAL_RW_DATA 5 [11:0]											

PORT1 LOCAL-RW DATA 6 ~ PORT8 LOCAL RW DATA 6

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h98~'h9F	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 6 [11:0] ~ PORT8_LOCAL_RW_DATA 6 [11:0]											

PORT1 LOCAL-RW DATA 7 ~ PORT8 LOCAL RW DATA 7

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hA0~'hA7	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 7 [11:0] ~ PORT8_LOCAL_RW_DATA 7 [11:0]											

PORT1 LOCAL-RW DATA 8 ~ PORT8 LOCAL RW DATA 8

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hA8~'hAF	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 8 [11:0] ~ PORT8_LOCAL_RW_DATA 8 [11:0]											

PORT1 LOCAL-RW DATA 9 ~ PORT8 LOCAL RW DATA 9

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hB0~'hB7	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 9 [11:0] ~ PORT8_LOCAL_RW_DATA 9 [11:0]											

PORT1 LOCAL-RW DATA 10 ~ PORT8 LOCAL RW DATA 10

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hB8~'hBF	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 10 [11:0] ~ PORT8_LOCAL_RW_DATA 10 [11:0]											

PORT1 LOCAL-RW DATA 11 ~ PORT8 LOCAL RW DATA 11

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hC0~'hC7	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 11 [11:0] ~ PORT8_LOCAL_RW_DATA 11 [11:0]											

PORT1 LOCAL-RW DATA 12 ~ PORT8 LOCAL RW DATA 12

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hC8~'hCF	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 12 [11:0] ~ PORT8_LOCAL_RW_DATA 12 [11:0]											

PORT1 LOCAL-RW DATA 13 ~ PORT8 LOCAL RW DATA 13

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hD0~'hD7	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 13 [11:0] ~ PORT8_LOCAL_RW_DATA 13 [11:0]											

PORT1 LOCAL-RW DATA 14 ~ PORT8 LOCAL RW DATA 14

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hD8~'hDF	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 14 [11:0] ~ PORT8_LOCAL_RW_DATA 14 [11:0]											

PORT1 LOCAL-RW DATA 15 ~ PORT8 LOCAL RW DATA 15

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hE0~'hE7	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 15 [11:0] ~ PORT8_LOCAL_RW_DATA 15 [11:0]											

PORT1 LOCAL-RW DATA 16 ~ PORT8 LOCAL RW DATA 16

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hE8~'hEF	0x0000	x	x	x	x	PORT1_LOCAL_RW_DATA 16 [11:0] ~ PORT8_LOCAL_RW_DATA 16 [11:0]											

Bits	Fields	Descriptions
11:0	PORT _x _LOCAL_RW_DATA _y	These registers are used to contain the data to be written/read to/from the XD12R drivers daisied to each channel for accessing the XD12 drivers' registers. These registers should be written before executing the relevant commands, the local-reading or local-writing command.

It needs to be careful to use that these registers are differently mapped according to the CH_SEG [1:0] of the LOCAL WRITE/READ COMMAND registers

CH_SEG = 2'b00 : PORT1 to PORT8 are mapped to the channel 1 to 8 in order.
 CH_SEG = 2'b01 : PORT1 to PORT8 are mapped to the channel 9 to 16 in order.
 CH_SEG = 2'b10 : PORT1 to PORT8 are mapped to the channel 17 to 24 in order.

The data order to be transferred to the XD12R drivers daisied in a channel_x
 : PORT_x_LOCAL_RW_DATA₁ to the 1st driver,
 PORT_x_LOCAL_RW_DATA₂ to the 2nd driver,

The data order to be received from the XD12R drivers daisied in a channel_x
 : PORT_x_LOCAL_RW_DATA₁ from the last driver,
 PORT_x_LOCAL_RW_DATA₂ from the (last -1) driver,

Refer to the Fig 15. for detail.

REGISTER GROUP SEL

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'hF0	0x0000	RSV	ADDR_EXT	x	x	x	RSV	RSV	RSV	x	x	x	RSV	x	x	RSV	RSV

Bits	Fields	Descriptions
15	RSV	Used for test only
14	ADDR_EXT	This bit is for use extension register address group2. '0' : group1 '1' : group2 Please access registers with careful selection between two groups.
10	RSV	Used for test only
9	RSV	Used for test only
8	RSV	Used for test only
4	RSV	Used for test only
1	RSV	Used for test only
0	RSV	Used for test only

DAC1 FB VALID CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h00	0x0000	DAC1_FB_VALID_CNT[17:2]															

Bits	Fields	Descriptions
15:0	DAC1_FB_VALID_CNT	This is for test only. This value shows the counted value, MCLK x 4, for one frame.

DAC1 INC HOLD/WAIT CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h01	0x0000	DAC1_INC_HOLD_LIMIT_CNT[8:0]								DAC1_INC1_WAIT_CNT[6:0]							

Bits	Fields	Descriptions
15:7	DAC1_INC_HOLD_LIMIT_CNT	This is for test only.
6:0	DAC1_INC1_WAIT_CNT	This is for test only.

1R2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h02	0x0000	x	x	x	SIGNED	IR2[11:0]											

Bits	Fields	Descriptions
12	SIGNED	The signed bit of the R2. This is for test only.
11:0	IR2	= (CURR_TGT_DAC – the current DAC)

SOA1_N11 & N1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h03	0x1908	SOA_N11 [8:0]										SOA_N1 [6:0]					

Bits	Fields	Descriptions
15:7	SOA_N11	This determines the condition whether the DAC OUT will be changed to the CURR_TGT_DAC or kept to the current DAC value in case that the condition $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N11}) < 0$ is met. If the current DAC is smaller than the CURR_TGT_DAC, then the DAC OUT will be kept to the current DAC, and if not so, the current DAC will be changed to the CURR_TGT_DAC value. Normally, this function will work when the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred while the DAC OUT is increasing.
6:0	SOA_N1	This determines the condition whether the DAC OUT will be immediately decreased by the amount of $(\text{CURR_TGT_DAC} - \text{the current DAC})$ or not. Whenever the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred, the DAC OUT will be immediately decreased by the amount of $(\text{CURR_TGT_DAC} - \text{the current DAC})$. Normally, this is working that the current frame's target is less than the previous frame's target by the SOA_N1. Refer to the Figure 28.

SOA1_P2 & P1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h04	0xC07D	SOA_P2 [3:0]				x	x	x	SOA1_P1 [8:0]								

Bits	Fields	Descriptions
15:12	SOA_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
8:0	SOA1_P1	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

SOA1_P3 & P2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h05	0x7D12	SOA1_P3 [9:0]										SOA1_P2 [9:4]					

Bits	Fields	Descriptions
		It must be careful that this register is prohibited from being written.
15:6	SOA1_P3	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
5:0	SOA1_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

DAC2 FB VALID CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h06	0x0000	DAC2_FB_VALID_CNT[17:2]															

Bits	Fields	Descriptions
15:0	DAC2_FB_VALID_CNT	This is for test only. This value shows the counted value, MCLK x 4, for one frame.

DAC2 INC HOLD/WAIT CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h07	0x0000	DAC2_INC_HOLD_LIMIT_CNT[8:0]										DAC2_INC1_WAIT_CNT[6:0]					

Bits	Fields	Descriptions
15:7	DAC2_INC_HOLD_LIMIT_CNT	This is for test only.
6:0	DAC2_INC1_WAIT_CNT	This is for test only.

2R2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h08	0x0000	x	x	x	SIGNED	2R2[11:0]											

Bits	Fields	Descriptions
12	SIGNED	The signed bit of the R2. This is for test only.
11:0	2R2	= (CURR_TGT_DAC – the current DAC)

SOA2_N11 & N1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h09	0x1908	SOA2_N11 [8:0]										SOA2_N1 [6:0]					

Bits	Fields	Descriptions
15:7	SOA2_N11	This determines the condition whether the DAC OUT will be changed to the CURR_TGT_DAC or kept to the current DAC value in case that the condition $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N11}) < 0$ is met. If the current DAC is smaller than the CURR_TGT_DAC, then the DAC OUT will be kept to the current DAC, and if not so, the current DAC will be changed to the CURR_TGT_DAC value. Normally, this function will work when the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred while the DAC OUT is increasing.
6:0	SOA2_N1	This determines the condition whether the DAC OUT will be immediately decreased by the amount of (CURR_TGT_DAC – the current DAC) or not. Whenever the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred, the DAC OUT will be immediately decreased by the amount of (CURR_TGT_DAC – the current DAC). Normally, this is working that the current frame's target is less than the previous frame's target by the SOA_N1. Refer to the Figure 28.

SOA2_P2 & P1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0A	0xC07D	SOA2_P2 [3:0]				x	x	x	SOA2_P1 [8:0]								

Bits	Fields	Descriptions
15:12	SOA2_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
8:0	SOA2_P1	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

SOA2_P3 & P2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0B	0x7D12	SOA2_P3 [9:0]										SOA2_P2 [9:4]					

Bits	Fields	Descriptions
		It must be careful that this register is prohibited from being written.
15:6	SOA2_P3	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
5:0	SOA2_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

DAC3 FB VALID CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0C	0x0000	DAC3_FB_VALID_CNT[17:2]															

Bits	Fields	Descriptions
15:0	DAC3_FB_VALID_CNT	This is for test only. This value shows the counted value, MCLK x 4, for one frame.

DAC3 INC HOLD/WAIT CNT

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0D	0x0000	DAC3_INC_HOLD_LIMIT_CNT[8:0]										DAC3_INC1_WAIT_CNT[6:0]					

Bits	Fields	Descriptions
15:7	DAC3_INC_HOLD_LIMIT_CNT	This is for test only.
6:0	DAC3_INC1_WAIT_CNT	This is for test only.

3R2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0E	0x0000	x	x	x	SIGNED	3R2[11:0]											

Bits	Fields	Descriptions
12	SIGNED	The signed bit of the R2. This is for test only.
11:0	3R2	= (CURR_TGT_DAC – the current DAC)

SOA3_N11 & N1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h0F	0x1908	SOA3_N11 [8:0]										SOA3_N1 [6:0]					

Bits	Fields	Descriptions
15:7	SOA3_N11	This determines the condition whether the DAC OUT will be changed to the CURR_TGT_DAC or kept to the current DAC value in case that the condition $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N11}) < 0$ is met. If the current DAC is smaller than the CURR_TGT_DAC, then the DAC OUT will be kept to the current DAC, and if not so, the current DAC will be changed to the CURR_TGT_DAC value. Normally, this function will work when the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred while the DAC OUT is increasing.
6:0	SOA3_N1	This determines the condition whether the DAC OUT will be immediately decreased by the amount of (CURR_TGT_DAC – the current DAC) or not. Whenever the condition, $((\text{CURR_TGT_DAC} - \text{PREV_TGT_DAC}) + \text{SOA_N1}) < 0$, is occurred, the DAC OUT will be immediately decreased by the amount of (CURR_TGT_DAC – the current DAC). Normally, this is working that the current frame's target is less than the previous frame's target by the SOA_N1. Refer to the Figure 28.

SOA3_P2 & P1

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h10	0xC07D	SOA3_P2 [3:0]				x	x	x	SOA3_P1 [8:0]								

Bits	Fields	Descriptions
15:12	SOA3_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
8:0	SOA3_P1	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

SOA3_P3 & P2

Addr	Default	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
'h11	0x7D12	SOA3_P3 [9:0]										SOA3_P2 [9:4]					

Bits	Fields	Descriptions
		It must be careful that this register is prohibited from being written.
15:6	SOA3_P3	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.
5:0	SOA3_P2	This determines the extent to which the DAC OUT will be increased. Refer to the DAC INCREMENT 1 & 2 registers and Figure 28. in details.

Soldering Temperature Reflow Profile

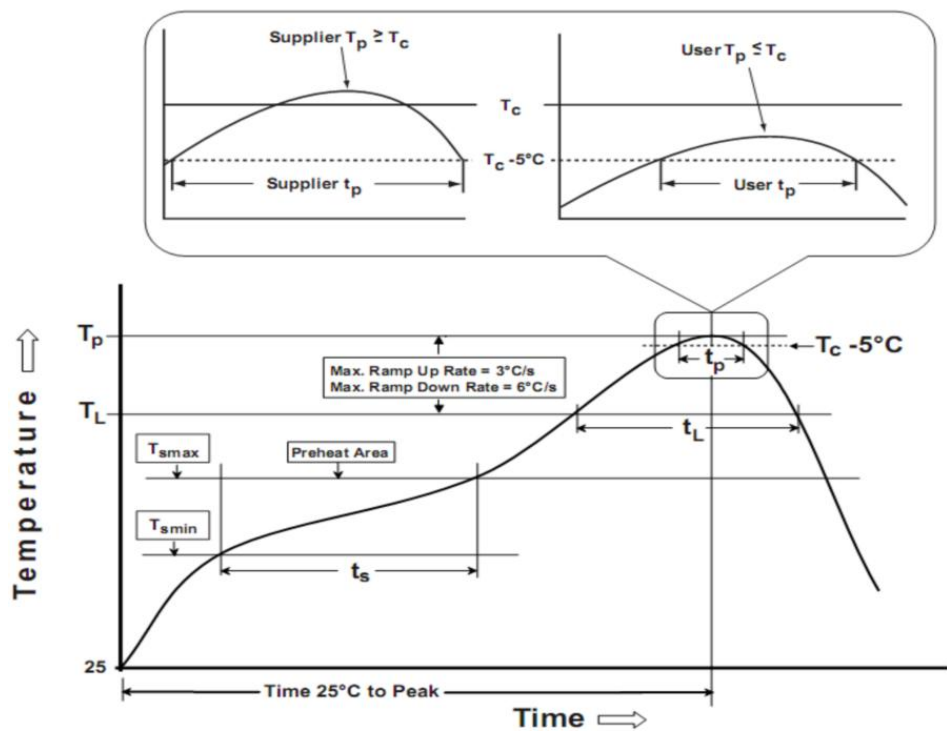


Figure 29. Reflow Soldering information

Parameter	Pb-free soldering conditions
Average ramp-up rate (217 °C to Peak)	3°C/second max.
Time of Preheat temp (from 150 °C to 200 °C)	60 ~ 120 second
Time to be maintained above 217 °C	60 ~ 150 second
Peak temperature	260 °C
Time within 5 °C of actual peak temp	30 second
Ramp-down rate	6 °C/second max.
Time from 25°C to peak temp	8 minutes max.

Table 10. Reflow Soldering Parameter

Note:

1. The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”
2. Tolerance for peak profile temperature (Tp) is defined as a supplier minimum and a user maximum.
3. This document should serve as recommendation only. Other parameters may also affect soldering, so these profiles do not guarantee absolute success.
4. Soldering profiles should be determined by reference to the solder and solder paste manufacturer's specifications, unless they conflict with the recommendations in this document.
5. The soldering profile and footprint information specified in this datasheet are general recommendations and are provided for reference only. The design, control, and quality assurance of the actual soldering process are the sole responsibility of the customer. The supplier assumes no responsibility for the soldering process or its results.

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

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Body TOP Marking

< JSICAT marking >



TOP VIEW



Fix. Code IC Part Name: XC24R

Package Drawings Markings

Row No.	Marking	Definition
1	XXXXX	Fixed Code: XC24R
2	abc	Internal code
3	YWCD@	Y: Year Code W: Week Code C: Wafer Code D: Traceability Wafer lot Code @: Assembly plant Code

** Font Type : Arial
Marking text tolerance : ±0.05mm

Assembly production Code : **CBA2J**

※ Example code :2025. 07. 07
wafer Lot #6-10

Wafer Lot	2024		2025		2026		2027		...	
	1-6	7-12	1-6	7-12	1-6	7-12	1-6	7-12	1-6	7-12
Code	-	A	B	C	D	E	F	G	...	...

W → Week code , A=W1, B=W2,...

Wafer Lot	W 1	W 2	W3	W4	...	W27	W28	...	W53
Code	A	B	C	D	...	A	B	...	A

C → Wafer lot, Each wafer Lot is represented by a character (Ex: A, B, C, ...).
These character cycle once a week.

Wafer Lot	Lot 1	Lot 2	Lot 3	Lot 4	Lot 5	Lot 6	Lot 7	Lot 8	...
Code	A	B	C	D	E	F	G	H	...

D → Traceability Code Wafer lot NO. :
A big assembly Lot consists of three wafers.
Each big assembly Lot.
Lot is represented by a character (Ex: 1,2,3,...)

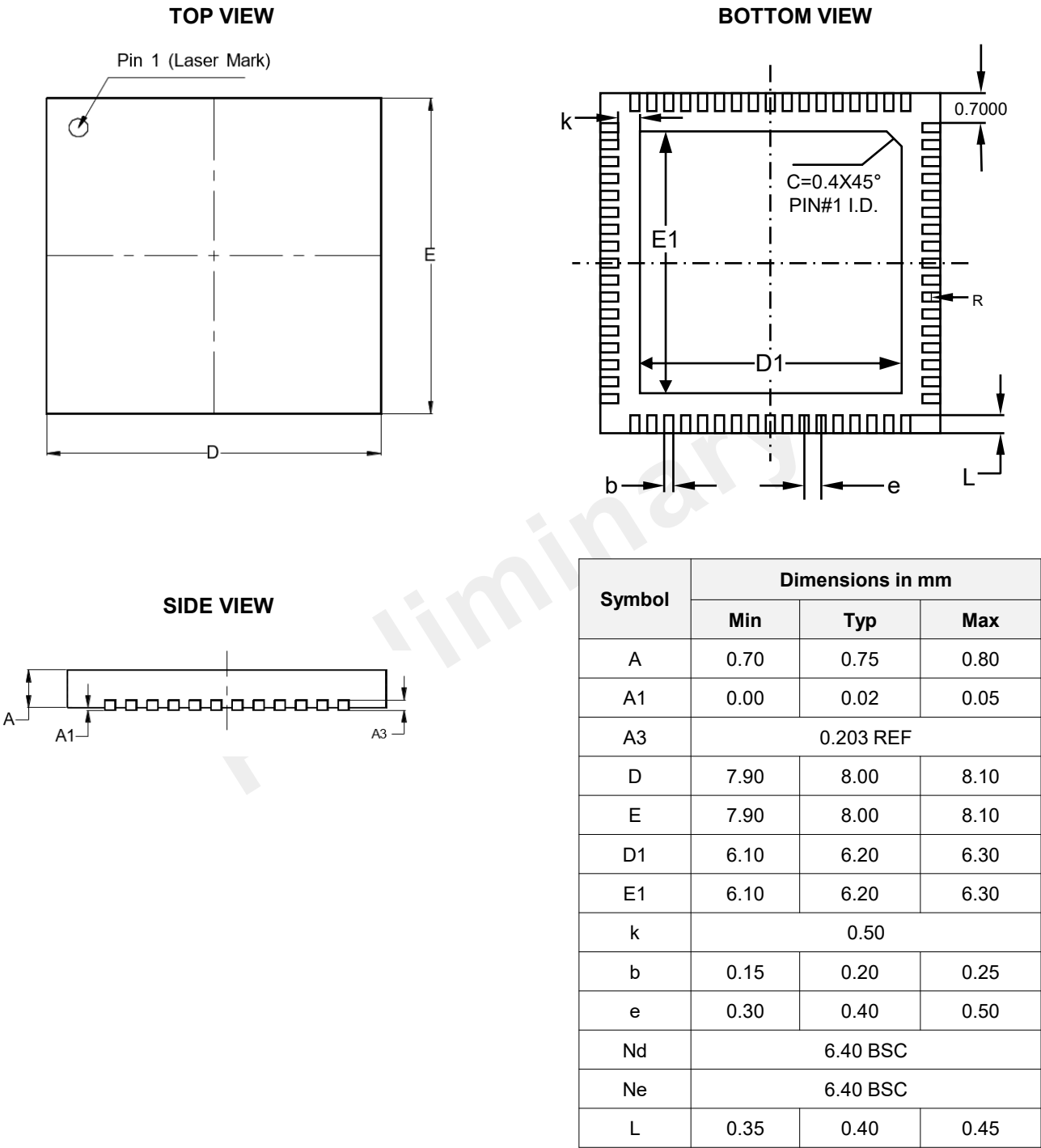
Wafer Lot No	1-5	6-10	11-15	16-20	21-25
Code	1	2	3	4	5

@ → Assembly plant code				
	JSICAT	SEMTECH	3 rd plant	4 th Plant
Code	J	S	*	*

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Package specification

QFN 8080-68L Package Outline

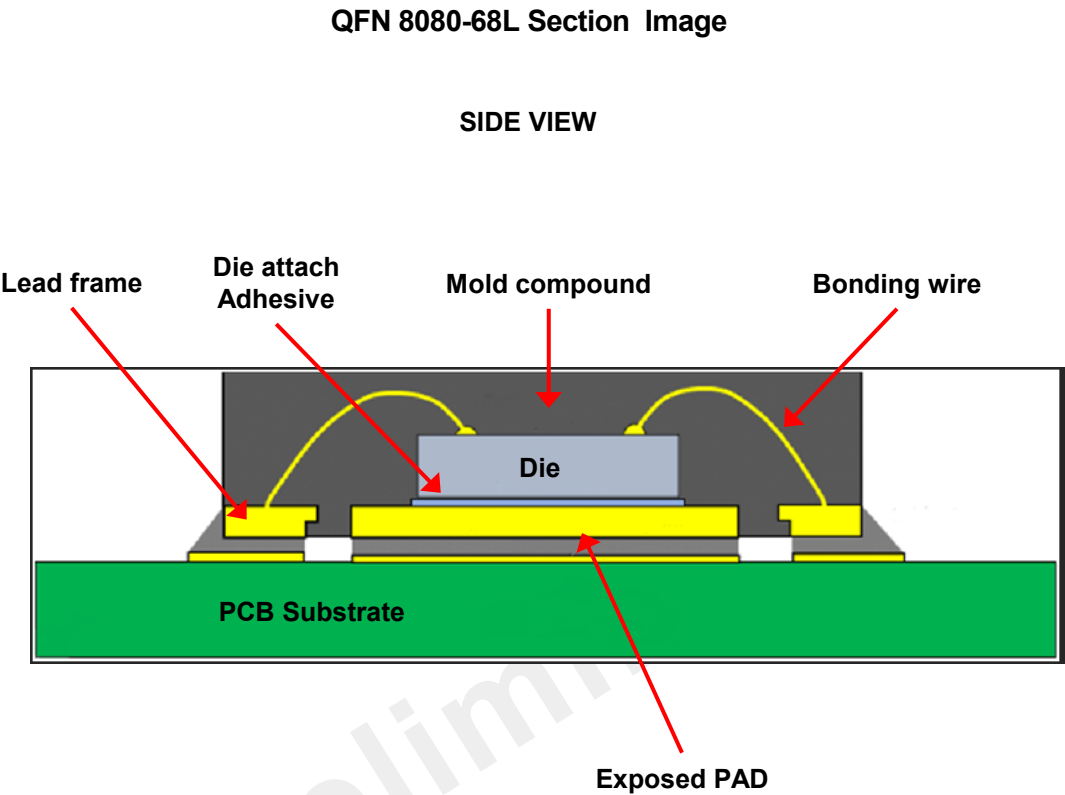


Note
All Dimensions Do Not Include mold flash or Protrusions.

Figure 32. Package Outline

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QFN 8080-68L Section Image



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Footprint

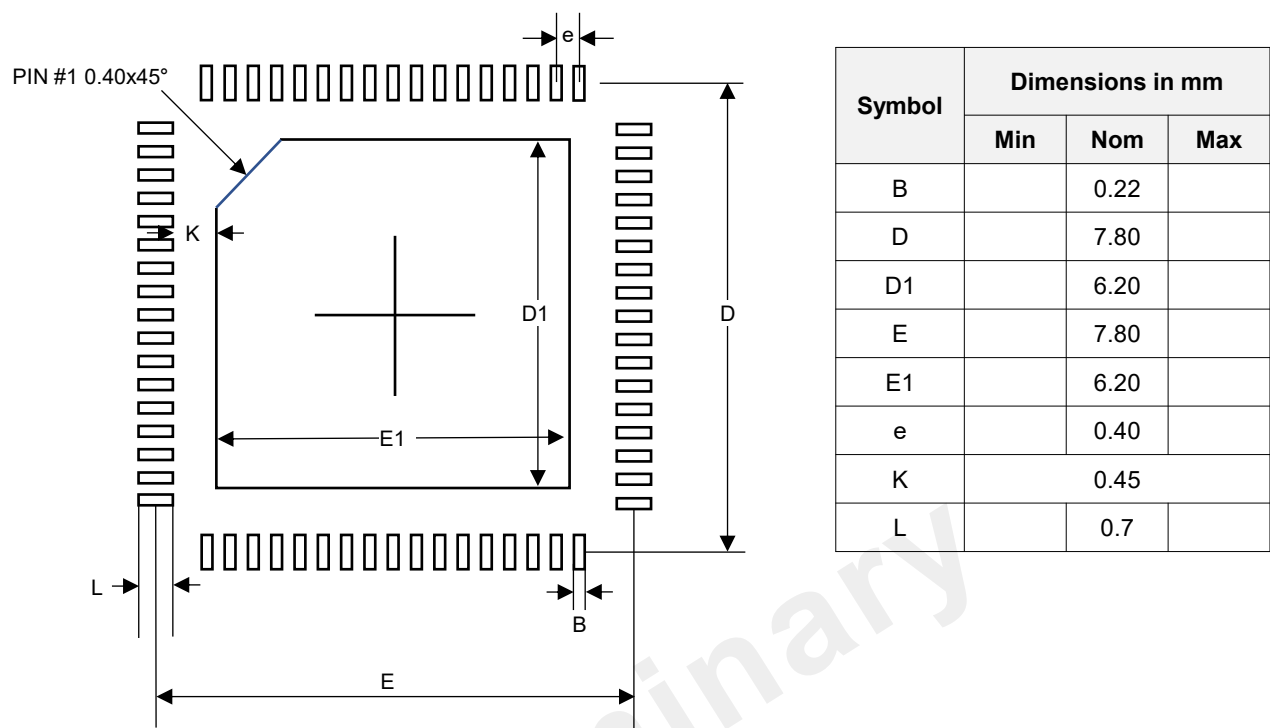


Figure 33. Recommended Footprint

Note : The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”

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